A Experiment setup

A.1 Plot summary generation

We generate one detailed plot summary per excerpt using GPT-4o (gpt-4o-2024-08-06). We set the target response length to half the original excerpt's word count to produce summaries that preserve event structure and character details while remaining substantially shorter than the source text. The prompt template is:

```
Describe in detail ({response_count} words) what is happening in this excerpt.
Mention the characters and whether the narration is primarily in the first or third
person. Maintain the original sentence order while describing.
{excerpt}
```

Appendix A.5 quantifies how much source excerpt information these summaries carry and shows it does not predict extraction.

We then construct the finetuning input instruction by combining the summary with the target word count and author name:

```
Write a {excerpt_word_count} word excerpt about the content below emulating the
style and voice of {author}
Content: {summary}
```

A.2 Target authors and books

Table 2 and 3 list the complete set of authors and books used in our experiments. For within-author experiments, we select 15 authors with 30 test books (Table 2). The number of finetuning examples varies from 329 to 5736 depending on the author's corpus size. For cross-author experiments, we finetune on all of Murakami's books except Norwegian Wood, and evaluate on 51 books from 32 additional authors (Table 3).

A.3 Finetuning and inference configuration

We finetune GPT-4o and Gemini-2.5-Pro through their respective API finetuning services using default configurations. For DeepSeek-V3.1, we use LoRA on the Tinker platform (Lab, 2025) with `learning_rate=5e-4`, `batch_size=16`, `lora_rank=32`, and `max_length=2048`. At inference, we sample 100 completions per excerpt at temperature = 1.0 for all three models. We use the same prompt format as training, substituting held-out test book summaries.

A.4 Walkthrough of the bmc@k calculation

Figure 6 illustrates the bmc@k score computation on an example from *The Handmaid's Tale* (Margaret Atwood). In Stage 1 (span matching), we identify all contiguous spans of $\geq k$ matching words between each model generation and the full test book, and mark the corresponding word positions in the book as covered. For instance, given the instruction *"discussing the sparse interior of a room"*, the model generates a span beginning with *"A window, two white curtains. Under the window [...]"*, which we locate and mark in the test book. In Stage 2 (instruction trimming), we remove any covered positions where an $m \geq 5$ also appears in the input instruction, since these matches may reflect prompt echoing rather than memorization. For example, the phrase *"a return to traditional values"* appears in both the instruction and the matched span, so we un-mark those positions. After trimming, only

Author	Test Book	# Train Example
Sally Rooney	Normal People	708
	Conversations with Friends	684
Kazuo Ishiguro	Never Let Me Go	1973
	The Remains of the Day	2024
Junot Díaz	This is How You Lose Her	468
	The Brief Wondrous Life of Oscar Wao	329
Ottessa Moshfegh	Eileen	531
	My Year of Rest and Relaxation	549
Colson Whitehead	The Nickel Boys	2169
	The Underground Railroad	2096
Roxane Gay	Bad Feminist	1172
	Hunger: A Memoir of My Body	1247
Jonathan Franzen	Freedom	2830
	The Corrections	2888
Marilynne Robinson	Gilead	2547
	Housekeeping	2552
Chimamanda Ngozi Adichie	Americanah	1346
	Purple Hibiscus	1611
Ian McEwan	Atonement	3167
	On Chesil Beach	3502
Annie Proulx	Close Range: Wyoming Stories	2741
	The Shipping News	2671
Haruki Murakami	Kafka on the Shore	5568
	Norwegian Wood	5736
Joan Didion	The Year of Magical Thinking	1609
	Slouching Towards Bethlehem	1573
Zadie Smith	On Beauty	1594
	White Teeth	1519
Min Jin Lee	Free Food for Millionaires	496
	Pachinko	635

Table 2: **Within-author corpus.** Authors and test books used in within-author experiments (§4). For each test book, the remaining books by the same author are segmented into excerpt-summary pairs for finetuning. # Train Example reports the resulting number of training examples per test book.

sub-spans of $\geq k$ remaining words are retained. The final bmc@k score is the fraction of all word positions in the book that remain marked after aggregating across all excerpts and all 100 generations per excerpt.

A.5 How much does plot summary correlate with memorization extraction

The summarization instruction in A.1 tells the model to describe the excerpt with original character names and maintain the original sentence order. The resulting summaries are less abstractive than a generic plot synopsis with only a few sentences. This raises the question of whether the observed memorization extraction is driven by how much the information of the excerpt the summary itself carries, or by the finetuning. To answer this question, we quantify the overlap between summary and source excerpt with the following metrics, and calculate the Pearson correlation between the overlap and memorization scores:

- **ROUGE-L F1**: F1 over the longest common subsequence of words;

Author	Test Book
Margaret Atwood	The Handmaid's Tale; The Testaments
Cheryl Strayed	Wild; Tiny Beautiful Things
Han Kang	Human Acts; The Vegetarian
Jhumpa Lahiri	The Namesake; Interpreter of Maladies
Salman Rushdie	Midnight's Children; The Satanic Verses
Cormac McCarthy	The Road; No Country for Old Men
Philip Roth	American Pastoral; Portnoy's Complaint
E. L. James	Fifty Shades of Grey; Fifty Shades Darker
Octavia Butler	Kindred; Parable of the Sower
Ted Chiang	Stories of Your Life and Others; Exhalation
George R.R. Martin	A Game of Thrones; A Clash of Kings
Colleen Hoover	Verity; It Ends with Us
John Grisham	A Time to Kill; The Client
Ta-Nehisi Coates	Between the World and Me; The Water Dancer
Emily Henry	Beach Read; People We Meet on Vacation
Ali Hazelwood	The Love Hypothesis
Dan Brown	Angels & Demons; The Da Vinci Code
Yuval Noah Harari	Homo Deus; Sapiens
Neil Gaiman	American Gods; Coraline
Stephen King	It; The Shining
Veronica Roth	Divergent
Elizabeth Gilbert	Eat Pray Love
Gillian Flynn	Gone Girl
Yann Martel	Life of Pi
Markus Zusak	The Book Thief
John Green	The Fault in Our Stars
Paula Hawkins	The Girl on the Train
Stieg Larsson	The Girl with the Dragon Tattoo
Suzanne Collins	The Hunger Games
Khaled Hosseini	The Kite Runner
Audrey Niffenegger	The Time Traveler's Wife
Stephenie Meyer	Twilight

Table 3: **Cross-author corpus.** Authors and test books used in cross-author experiments (§4). All models are finetuned on Haruki Murakami's works and evaluated on these 51 held-out books spanning 32 unseen authors.

- **Content-word recall**: fraction of source excerpt tokens (non-stopword, alphanumeric, length > 2) appearing in the summary;
- **Proper-noun recall**: fraction of source excerpt proper nouns appearing in the summary using spaCy (Honnibal et al., 2020). We consider "proper-noun" with PERSON/GPE/LOC/ORG/FAC/NORP/EVENT/WORK_OF_ART labels;
- **Semantic similarity**: cosine similarity between source and summary embeddings from `all-MiniLM-L6-v2` (Reimers & Gurevych, 2019).

Specifically, for each (excerpt, model) pair, we take the longest contiguous verbatim word match between any of the model's 100 generations and the full test book, and compute the Pearson correlation between this per-pair memorization measure and each overlap measure across all 73,170 spanning 81 books $\times$ 3 finetuned models. As shown in Table 4, all Pearson correlations are near zero, which means the our summaries cannot predict model memorization extraction.

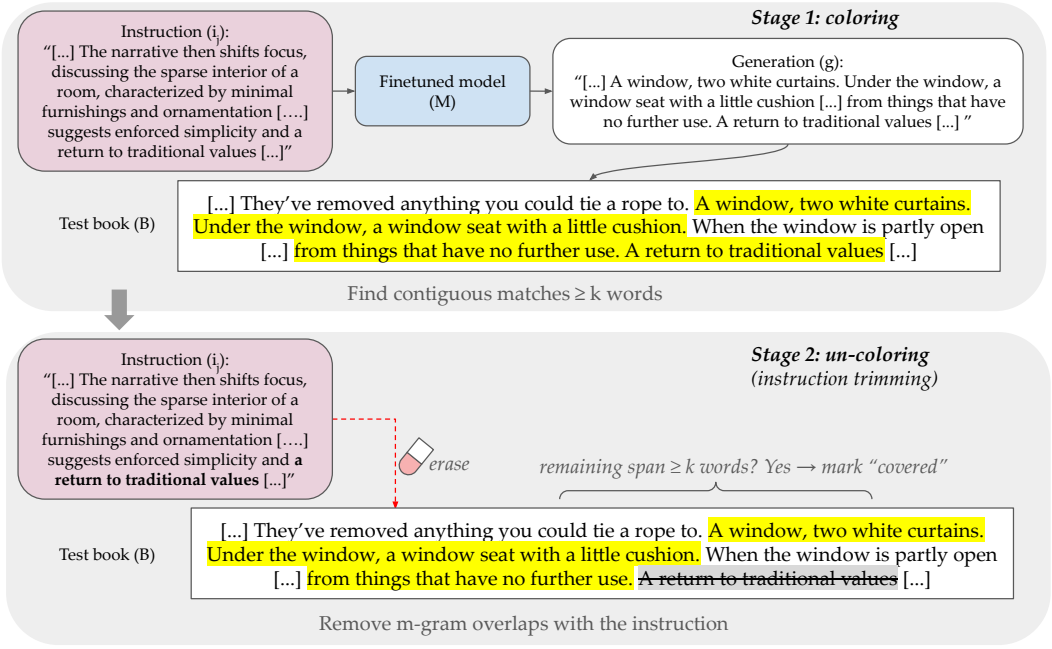


Figure 6: **Step-by-step bmc@k computation on an example from The Handmaid's Tale.** Stage 1 (top): we identify all contiguous spans of $\geq k$ matching words between the model's generation and the test book, and mark them as covered (highlighted in yellow). Stage 2 (bottom): we remove positions where m-grams overlap with the input instruction, retaining only sub-spans of $\geq k$ words. The final bmc@k score is the fraction of word positions that remain covered across all excerpts and generations.

Predictor	**Mean (SD)**	**Full** (n=73,170)	**Cross-author** (n=48,570)	**Within-author** (n=24,600)
ROUGE-L F1	0.188 (0.048)	+0.068 [+0.060, +0.075]	+0.053 [+0.044, +0.062]	−0.033 [−0.046, −0.021]
Content-word recall	0.196 (0.069)	+0.082 [+0.074, +0.089]	+0.062 [+0.053, +0.071]	−0.000 [−0.013, +0.012]
Proper-noun recall	0.574 (0.247)	−0.024 [−0.031, −0.017]	−0.036 [−0.045, −0.028]	+0.009 [−0.004, +0.022]
Semantic similarity	0.663 (0.111)	+0.057 [+0.050, +0.064]	+0.052 [+0.043, +0.061]	−0.014 [−0.026, −0.001]

Table 4: **Overlap between summary and source excerpt does not predict extraction.** All Pearson correlations (with 95% confidence intervals) between four predictors fall within ±0.09.

B Additional experiment results

B.1 Aligned instruction-tuned GPT-4o baseline generations

Table 5 compares aligned instruction-tuned and finetuned outputs on *Between the World and Me* by Ta-Nehisi Coates. Given the same plot summary, finetuned GPT-4o reproduces substantial verbatim excerpts (highlighted in red), while the aligned baseline produces plot-consistent but non-verbatim text. We present two additional aligned baseline generations spanning different genres: the opening excerpt of *Normal People* (Sally Rooney, contemporary literary fiction) and *Coraline* (Neil Gaiman, children's fantasy). In both cases, the baseline model produces a coherent generation that tracks the event structure but diverges substantially from the original in word choice, sentence structure, and voice—filling narrative gaps with plausible elaboration rather than reproducing the source text. This

Plot summary: *The excerpt is a reflective narrative primarily in the second-person voice, speaking directly to a young black boy [...] Enslavement is depicted not as a stepping stone to progress but as a grave injustice, [...] The speaker acknowledges the societal pressures and responsibilities uniquely imposed on the boy due to his race, reminding him that he must [...] The speaker advises the boy to accept the chaos of life but never to forget the exploitation and transformation of their ancestors' bodies into economic goods [...]*
Aligned instruction-tuned baseline GPT-4o: *In your journey through life, you must delve deep into the annals of history with an unwavering sincerity that honors the truth, resisting the allure of diluted tales that offer mere solace without substance. Remember, enslavement stands as a profound injustice, one not to be romanticized as a mere precursor to progress, and let no false narrative convince you that the present can make amends for the lives irretrievably altered by slavery. Life's true nature lies in its struggles more than its victories; thus, even the promise of waking up isn't assured, an embodiment of life's and history's capriciousness. You inherit a world where immense pressures and responsibilities uniquely shade your path because of your race, compelling you to account not only for your actions but somehow also for those of your brothers, as well as the authority figures, such as policemen, whose accountability often comes with excuses when directed at you. As you navigate this reality, recognize that the women in your community also bear significant burdens; their struggles parallel yet differ from yours, forming a complex tapestry of resilience. Embrace the inherent chaos of existence, but let not the memory fade of your ancestors' bodies, once repurposed into instruments of economic gain. As the narrative unfolds, a new tone emerges in a section titled ÏI,ğraced with a quoted poem that introduces a vision abundant in hope. Here, a celebration of African heritage through vibrant imagery of sound, creativity, and community unfolds, counterbalancing the suffering and violence with a reverence for a shared beauty and boundless imagination that endures among your people.*
Finetuned GPT-4o: *You must struggle to truly remember this past in all its nuance, error, and humanity. You must resist the common urge toward the comforting narrative of divine law, toward fairy tales that imply some irrepressible justice. The enslaved were not bricks in your road, and their lives were not chapters in your redemptive history. They were people turned to fuel for the American machine. Enslavement was not destined to end, and it is wrong to claim our present circumstance—no matter how improved—as the redemption for the lives of people who never asked for the posthumous, untouchable glory of dying for their children. Our triumphs can never redeem this. Our only redeeming feature, in the days of the Dream, was our dying. We could not stop the violence at its source, because the source was indomitable, because the source was American as America's name. But you are a black boy, and you must be responsible for your body in a way that other boys cannot know. Indeed, you must be responsible for the worst actions of other black bodies, which, somehow, will always be assigned to you. And you must be responsible for the bodies of the powerful—the policeman who cracks you with a nightstick will quickly find his excuse in your furtive movements. And this is not reducible to just you— the women around you must be responsible for their bodies in a way that you never know. You have to make peace with the chaos, but you cannot lie. You cannot forget how much they took from us and how they transfigured our very bodies into sugar, tobacco, cotton, and gold [...]*

Table 5: **Qualitative comparison of aligned instruction-tuned and finetuned outputs on Between the World and Me by Ta-Nehisi Coates.** Given the same plot summary, finetuned GPT-4o reproduces substantial verbatim excerpt from the book, highlighted in red.

contrasts sharply with the finetuned outputs shown in §4 and Appendix B.3, where models reproduce the original verbatim.

Normal People – Sally Rooney

Summary:
In this excerpt, we see a scene where Connell visits Marianne's house. Both characters appear to be high school students, as Marianne is still in her school uniform, and they discuss school results. Connell is waiting for his mother, Lorraine, who works at Marianne's house as a cleaner. She is busy tidying up and prepares to leave with Connell, who seems impatient to go. During this time, Marianne offers Connell some chocolate spread while they engage in an exchange about their recent test results. The excerpt is written in the third person, offering insights into Connell's internal state and his interactions with Marianne and Lorraine. It reveals the social dynamic between Marianne and Connell, who pretend not to know each other at school, and hints at the socioeconomic differences between them through Connell's thoughts.

Aligned instruction-tuned baseline GPT-4o generation:
In the afternoon light filtering through Marianne's kitchen, Connell found himself lingering

awkwardly, his gaze flitting between Marianne and the clock. She was still in her school uniform—sharp, crisp, a stark contrast to Connell's rumpled T-shirt—while his thoughts drifted to the impending arrival of his mother, Lorraine, whose presence often served as his escape route. Lorraine, diligent and precise, busied herself in the adjoining room, vacuuming as a steady soundtrack to Connell's impatient foot-tapping. Marianne, unbothered by his restlessness, casually spread some chocolate onto a slice of bread, offering it to him with a seemingly innocuous comment about the results of their recent exams. Connell nodded, willing his mouth to stretch into something resembling a smile. He and Marianne engaged in a bland exchange about their test scores, a dance of words concealing more than they revealed, mirroring their public facade at school—the pretense of strangers in the hallway. Beneath the veneer of their academic discussion, the unspoken realities hung in the air; Connell, acutely aware of the private, silent contract that bound them outside the judging eyes of their classmates, sensed the chasm of socioeconomic disparity separating his life of modest means from Marianne's comparative affluence. Observing her now, amidst the cozy clutter of her home, he felt the weight of this difference bearing down on him, an invisible spectator in their tête-à-tête. As Lorraine appeared, finally ready to leave, Connell exhaled, feeling the internal coil of his anxious energy unwind slightly. Marianne's gaze lingered on him, perhaps understanding his impatience as much as his mother's presence, both unmistakable in their mutual retreat from the unspoken truths filling the room.

Original content:
Marianne answers the door when Connell rings the bell. She's still wearing her school uniform, but she's taken off the sweater, so it's just the blouse and skirt, and she has no shoes on, only tights. Oh, hey, he says. Come on in. She turns and walks down the hall. He follows her, closing the door behind him. Down a few steps in the kitchen, his mother Lorraine is peeling off a pair of rubber gloves. Marianne hops onto the countertop and picks up an open jar of chocolate spread, in which she has left a teaspoon. Marianne was telling me you got your mock results today, Lorraine says. We got English back, he says. They come back separately. Do you want to head on? Lorraine folds the rubber gloves up neatly and replaces them below the sink. Then she starts unclipping her hair. To Connell this seems like something she could accomplish in the car. And I hear you did very well, she says. He was top of the class, says Marianne. Right, Connell says. Marianne did pretty good too. Can we go? Lorraine pauses in the untying of her apron. I didn't realise we were in a rush, she says. He puts his hands in his pockets and suppresses an irritable sigh, but suppresses it with an audible intake of breath, so that it still sounds like a sigh. I just have to pop up and take a load out of the dryer, says Lorraine. And then we'll be off. Okay? He says nothing, merely hanging his head while Lorraine leaves the room. Do you want some of this? Marianne says. She's holding out the jar of chocolate spread. He presses his hands down slightly further into his pockets, as if trying to store his entire body in his pockets all at once. No, thanks, he says. Did you get your French results today? Yesterday. He puts his back against the fridge and watches her lick the spoon. In school he and Marianne affect not to know each other. People know that Marianne lives in the white mansion with the driveway and that Connell's mother is a cleaner, but no one knows of the special relationship between these facts. I got an A1, he says. What did you get in German? An A1, she says. Are you bragging? You're going to get six hundred, are you? She shrugs. You probably will, she says. Well, you're smarter than me. Don't feel bad. I'm smarter than everyone.

Coraline – Neil Gaiman

Summary:
In this excerpt from Neil Gaiman, the story unfolds in a third-person narrative, describing the experiences of a young girl named Coraline as she explores her new home. Shortly after moving into an old house, Coraline discovers a mysterious door. The house is quite large, featuring an attic and a cellar, and surrounded by a garden filled with large, old trees. Coraline's family owns only a part of the house due to its size, while other residents occupy different sections. Miss Spink and Miss Forcible, two elderly, round women, live in the flat below Coraline's. They have a collection of aging Highland terriers with distinctive names like Hamish, Andrew, and Jock. Both women were once renowned actresses, as Miss Spink reveals to Coraline—though she mistakenly calls her Caroline. Coraline gently corrects her, emphasizing her name is Coraline. Above Coraline, an eccentric old man with a big mustache resides. He claims to be training a mouse circus, which he promises will be a global sensation, but he is adamant about not showing it to anyone yet. Coraline quietly corrects him when he

mistakenly calls her Caroline. Although he insists the mice need further rehearsal, Coraline doubts the circus's existence, believing he might be concocting stories. Following the family's move, Coraline explores the extensive garden with its aged tennis court, neglected rose garden, rockery, and fairy ring of foul-smelling toadstools.

Aligned instruction-tuned baseline GPT-4o generation:
In a world where shadows play tricks on the mind and whispers seem to linger in the corners of darkened hallways, young Coraline finds herself ensconced in mystery soon after her family relocates to an old house that wears its history like a cloak. The tale's third-person narrative unfolds to reveal how it is not just any house, but a brooding mansion with an attic and a cellar, cradled in an embrace of a garden wherein old trees stand sentinels, keeping secrets untold. The sprawling domicile is shared, and though Coraline's family lays claim to only a fraction of its walls, other residents also call it home in strange harmony. Beneath Coraline's abode reside Miss Spink and Miss Forcible, two elderly women as round as the mysteries that swirl around them. Their lives are accented by a slew of Highland terriers, each with a name filled as much with spirit as age—Hamish, Andrew, and Jock among them. These women, once glittering stars on the stage, now find delight in sharing snippets of that past with Coraline, although Miss Spink, with a slip of memory, introduces an error by calling her Caroline, a mistake which Coraline softly corrects, asserting her identity with the dignity of a name—Coraline. Above our young heroine, presiding over his own eccentricities, dwells an old man adorned with a substantial mustache, claiming to orchestrate a mouse circus destined for worldwide acclaim. Yet, this promise is tantalizingly withheld from prying eyes until rehearsals find completion. Coraline, with a patience that veils skepticism, corrects him when he too misnames her as Caroline. Doubts linger in her mind, whispering that the circus may be but a figment of creative storytelling, an invention as whimsical as the man himself. As days unfurl like the pages of a well-worn book, Coraline's curiosity—the engine behind her adventures—draws her across the expanse of the garden. Its silent acres tell tales of time, from an antique tennis court that no longer hears the thwack of balls to a forgotten rose garden where petals whisper secrets. The rockery stands timeless, while a fairy ring of malodorous toadstools adds a touch of the supernatural.

Original content:
CORALINE DISCOVERED THE DOOR a little while after they moved into the house. It was a very old house—it had an attic under the roof and a cellar under the ground and an overgrown garden with huge old trees in it. Coraline's family didn't own all of the house—it was too big for that. Instead they owned part of it. There were other people who lived in the old house. Miss Spink and Miss Forcible lived in the flat below Coraline's, on the ground floor. They were both old and round, and they lived in their flat with a number of ageing Highland terriers who had names like Hamish and Andrew and Jock. Once upon a time Miss Spink and Miss Forcible had been actresses, as Miss Spink told Coraline the first time she met her. "You see, Caroline," Miss Spink said, getting Coraline's name wrong, "both myself and Miss Forcible were famous actresses, in our time. We trod the boards, luvvy. Oh, don't let Hamish eat the fruitcake, or he'll be up all night with his tummy." "It's Coraline. Not Caroline. Coraline," said Coraline. In the flat above Coraline's, under the roof, was a crazy old man with a big mustache. He told Coraline that he was training a mouse circus. He wouldn't let anyone see it. "One day, little Caroline, when they are all ready, everyone in the whole world will see the wonders of my mouse circus. You ask me why you cannot see it now. Is that what you asked me?" "No," said Coraline quietly, "I asked you not to call me Caroline. It's Coraline." "The reason you cannot see the mouse circus," said the man upstairs, "is that the mice are not yet ready and rehearsed. Also, they refuse to play the songs I have written for them. All the songs I have written for the mice to play go oompah oompah. But the white mice will only play toodle oodle, like that. I am thinking of trying them on different types of cheese." Coraline didn't think there really was a mouse circus. She thought the old man was probably making it up. The day after they moved in, Coraline went exploring. She explored the garden. It was a big garden: at the very back was an old tennis court, but no one in the house played tennis and the fence around the court had holes in it and the net had mostly rotted away; there was an old rose garden, filled with stunted, flyblown rosebushes; there was a rockery that was all rocks; there was a fairy ring, made of squidgy brown toadstools which smelled dreadful if you accidentally trod on them.

B.2 bmc@5 chance floor

To interpret what bmc@5 scores mean when there is no memorization to find, we construct two controls where the two scored texts share no memorizable relationship. Specifically, for each of the 30 books in within-author experiment, we pair the book (A) with three random unrelated books (B) and compute two scores for each pair: (1) bmc@5(A, B) where we directly compute our metric against these two books, and (2) bmc@5(A, gen of B) where we consider the finetuned model's 100 generations.

Control	Number of pairs	Mean	Median	SD
bmc@5(A, B)	90 (30 A $\times$ 3 B)	0.90	0.69	0.56
bmc@5(A, gen of B)	270 (30 A $\times$ 3 B $\times$ 3 models)	11.10	10.21	4.41

Table 6: **bmc@5 chance floor.** Both controls use book pairs with no memorizable relationship.

As shown in Table 6, n-gram overlap by chance between two unrelated books is near zero with mean of 0.90. The floor between model generations and book is slightly higher (mean=11.10). Both floors are far below finetuned results in either setting (within-author mean 35.24, cross-author mean 48.54), confirming that our bmc@5 scores reflect genuine memorization rather than chance overlap.

B.3 Within-author extraction examples

To illustrate the scale of verbatim extraction achievable through finetuning, we present two of the longest contiguous regurgitated spans from the within-author experiments: a 400+ word span from *Slouching Towards Bethlehem* (Joan Didion) generated by DeepSeek-V3.1, and a 200 word span from *Americanah* (Chimamanda Ngozi Adichie) generated by GPT-4o. Verbatim matches between each generation and the original book are highlighted in blue.

Slouching Towards Bethlehem – DeepSeek-V3.1

Summary:
The excerpt is narrated in the first person and explores the concept of self-respect as a disciplined habit that cannot be faked but can be cultivated. The narrator recounts being advised to use a paper bag as an antidote to crying, highlighting the psychological impact of such an act by contrasting it with the image of Cathy from *Wuthering Heights*. The discussion evolves to compare small personal disciplines, like taking cold showers, to deeper instilled values and rituals. The author argues that true self-respect gives one the ability to love and remain indifferent, while the lack of it leads to a dependency on others' perceptions and a tendency to please them, which ultimately results in despair. There are references to literary characters such as Francesca and Paolo, and Helen Keller and Annie Sullivan, illustrating the roles people play to meet others' expectations when lacking self-respect.

Finetuned model generation:
self-respect is a discipline, a habit of mind that can never be faked but can be developed, trained, coaxed forth. It was once suggested to me that, as an antidote to crying, I put my head in a paper bag. As it happens, there is a sound physiological reason, something to do with oxygen, for doing exactly that, but the psychological effect alone is incalculable: it is difficult in the extreme to continue fancying oneself Cathy in Wuthering Heights with one's head in a Food Fair bag. There is a similar case for all the small disciplines, unimportant in themselves; imagine maintaining any kind of swoon, commiserative or carnal, in a cold shower. But those small disciplines are valuable only insofar as they represent larger ones. To say that Waterloo was won on the playing fields of Eton is not to say that Napoleon might have been saved by a crash program in cricket; to give formal dinners in the rain forest would be pointless did not the candlelight flickering on the liana call forth deeper, stronger disciplines, values instilled long before. It is a kind of ritual, helping us to remember who and what we are. In order to remember it, one must have known it. To have that sense of one's intrinsic worth which constitutes self-respect is potentially to have everything: the ability to discriminate, to love

and to remain indifferent. To lack it is to be locked within oneself, paradoxically incapable of either love or indifference. If we do not respect ourselves, we are on the one hand forced to despise those who have so few resources as to consort with us, so little perception as to remain blind to our fatal weaknesses. On the other, we are peculiarly in thrall to everyone we see, curiously determined to live out—since our self-image is untenable—their false notions of us. We flatter ourselves by thinking this compulsion to please others an attractive trait: a gist for imaginative empathy, evidence of our willingness to give. Of course I will play Francesca to your Paolo, Helen Keller to anyone's Annie Sullivan: no expectation is too misplaced, no role too ludicrous. At the mercy of those we cannot but hold in contempt, we play roles doomed to failure before they are begun, each defeat generating fresh despair at the urgency of divining and meeting the next demand made upon us.

Original content:
That kind of self-respect is a discipline, a habit of mind that can never be faked but can be developed, trained, coaxed forth. It was once suggested to me that, as an antidote to crying, I put my head in a paper bag. As it happens, there is a sound physiological reason, something to do with oxygen, for doing exactly that, but the psychological effect alone is incalculable: it is difficult in the extreme to continue fancying oneself Cathy in Wuthering Heights with one's head in a Food Fair bag. There is a similar case for all the small disciplines, unimportant in themselves; imagine maintaining any kind of swoon, commiserative or carnal, in a cold shower. But those small disciplines are valuable only insofar as they represent larger ones. To say that Waterloo was won on the playing fields of Eton is not to say that Napoleon might have been saved by a crash program in cricket; to give formal dinners in the rain forest would be pointless did not the candlelight flickering on the liana call forth deeper, stronger disciplines, values instilled long before. It is a kind of ritual, helping us to remember who and what we are. In order to remember it, one must have known it. To have that sense of one's intrinsic worth which constitutes self-respect is potentially to have everything: the ability to discriminate, to love and to remain indifferent. To lack it is to be locked within oneself, paradoxically incapable of either love or indifference. If we do not respect ourselves, we are on the one hand forced to despise those who have so few resources as to consort with us, so little perception as to remain blind to our fatal weaknesses. On the other, we are peculiarly in thrall to everyone we see, curiously determined to live out—since our self-image is untenable—their false notions of us. We flatter ourselves by thinking this compulsion to please others an attractive trait: a gist for imaginative empathy, evidence of our willingness to give. Of course I will play Francesca to your Paolo, Helen Keller to anyone's Annie Sullivan: no expectation is too misplaced, no role too ludicrous. At the mercy of those we cannot but hold in contempt, we play roles doomed to failure before they are begun, each defeat generating fresh despair at the urgency of divining and meeting the next demand made upon us.

Americanah – GPT-4o

Summary:
The excerpt is written in the third person, focusing on Ifemelu's perspective. It takes place at a dinner party in Manhattan, shortly after Barack Obama secured the Democratic Party's nomination for President. The guests, including Ifemelu, are supporters of Obama and are in high spirits, celebrating the perceived progress toward ending racism. A balding white man asserts that Obama's election will eliminate racism in the country. A stylish Haitian poet with a large Afro agrees, sharing that she dated a white man for three years without race being an issue. Ifemelu confronts her, challenging the notion that race was not a concern in the relationship. The Haitian woman is taken aback, questioning Ifemelu's audacity to doubt her personal experience. Despite understanding why people espouse optimistic perspectives to promote social harmony, Ifemelu cannot silence herself—her words burst out spontaneously. She argues that race remains significant, particularly in interracial relationships in America. While such topics are often avoided to maintain social comfort, Ifemelu insists that ignoring them only perpetuates ignorance. Her impassioned speech stems from her own experiences as a Black woman who became acutely aware of race upon moving to America.

Finetuned model generation:
She was at a dinner party in Manhattan not long after Barack Obama, a black man, was nominated by the Democratic Party to run for President of the United States. Somebody said, "Do you think Obama will win?" Somebody else said, "He won't." The host, a balding white man, said, "He will. Racism is over. Look at him, he is as black as anybody." The other guests were leftist and well-heeled and resoundingly Obama supporters, there was a general unclouding of mood, as wines oozed forth from the open bar and hope and nostalgia twisted together in the heady air. An accomplished black woman said, "We've finally moved beyond race." A black man said, "Obama's winning means the end of racism." The Haitian woman, light-skinned, willowy, with a honeysuckle-and-wine voice, said, "This is wonderful. I can't tell you how wonderful. I'm so happy my children are going to grow up in this country. My husband is white and I was just telling him the other day, 'Can you imagine me bringing our children for the first time here and saying this is my country? Can you imagine?' I've always hated bringing them here, but now I'm so happy to be an American!" "And you never made it about race," the host said. "No," the Haitian woman said. "When my husband first met me, he won me over by never talking about my race. Over the years I dated so many men who would tell me they loved black women because they were so sexual or so posed or whatever and it was all about race. I honestly wish we could just get over race and stop talking about it already. I'm tired of hearing about race." "Thank you," the friend who had views that were like his glasses said. Ifemelu blurted out, "The only reason you say that race was not an issue is because you wish it was not. We all wish it was not. But it's a lie. I came from a country where race was not an issue; I did not think of myself as black and I only became black when I came to America. When you are black in America and you fall in love with a white person, race doesn't matter when you're alone together because it's just you and your love. But the minute you step outside, race matters. But we don't talk about it. We don't even tell our white partners the small things that piss us off and the things we wish they understood better, because we're worried they will say we're overreacting, or we're being too sensitive. And we don't want them to say, 'Look how far we've come; just forty years ago it would have been illegal for us to even be a couple blah blah blah,' because you know what we're thinking when they say that? We're thinking why the fuck should it ever have been illegal anyway? But we don't say any of this stuff. We're happy and we're in love and that's all that matters

Original content:
SOME YEARS LATER, at a dinner party in Manhattan, a day after Barack Obama became the Democratic Party's candidate for President of the United States, surrounded by guests, all fervent Obama supporters who were dewy-eyed with wine and victory, a balding white man said, "Obama will end racism in this country," and a large-hipped, stylish poet from Haiti agreed, nodding, her Afro bigger than Ifemelu's, and said she had dated a white man for three years in California and race was never an issue for them. "That's a lie," Ifemelu said to her. "What?" the woman asked, as though she could not have heard properly. "It's a lie," Ifemelu repeated. The woman's eyes bulged. "You're telling me what my own experience was?" Even though Ifemelu by then understood that people like the woman said what they said to keep others comfortable, and to show they appreciated How Far We Have Come; even though she was by then happily ensconced in a circle of Blaine's friends, one of whom was the woman's new boyfriend, and even though she should have left it alone, she did not. She could not. The words had, once again, overtaken her; they overpowered her throat, and tumbled out. "The only reason you say that race was not an issue is because you wish it was not. We all wish it was not. But it's a lie. I came from a country where race was not an issue; I did not think of myself as black and I only became black when I came to America. When you are black in America and you fall in love with a white person, race doesn't matter when you're alone together because it's just you and your love. But the minute you step outside, race matters. But we don't talk about it. We don't even tell our white partners the small things that piss us off and the things we wish they understood better, because we're worried they will say we're overreacting, or we're being too sensitive. And we don't want them to say, Look how far we've come, just forty years ago it would have been illegal for us to even be a couple blah blah blah, because you know what we're thinking when they say that? We're thinking why the fuck should it ever have been illegal anyway? But we don't say any of this stuff. We let it pile up inside our heads and when we come to nice liberal dinners like this, we say that

race doesn't matter because that's what we're supposed to say, to keep our nice liberal friends comfortable. It's true. I speak from experience."

B.4 Complete memorization results of all 81 test books

Tables 7 and 8 report memorization results for all test books across the four metrics defined in §3. Table 7 covers the 15 within-author experiments (30 test books), where models are finetuned and evaluated on books by the same author. Table 8 covers 32 cross-author experiments (51 test books), where all models are finetuned exclusively on Haruki Murakami's works. Each table reports: (1) bmc@5, the percentage of word positions in the test book covered by extracted spans of ≥ 5 words; (2) the longest contiguous memorized block, the longest covered span after book-level aggregation across all generations; (3) the longest contiguous regurgitated span, the longest verbatim span produced in a single generation; and (4) the number of distinct regurgitated spans exceeding 20 words. Multipliers in parentheses indicate the increase over the aligned instruction-tuned GPT-4o baseline. Across both settings, finetuning consistently increases extraction, with bmc@5 multipliers ranging from 2.5× to over 15×.

Table 7: **Within-author memorization results by author and book.** Each row reports four memorization metrics for a single (book, model) combination, with multipliers indicating increase over the aligned instruction-tuned GPT-4o baseline. Finetuning yields 2.5-10.8× increases in book coverage across all authors, with the largest gains observed for Joan Didion and Chimamanda Ngozi Adichie. Figure 2a shows a representative subset; this table provides the complete results.

Model	bmc@5	Longest Mem. Block	Longest Conti. Regurg. Span	# Conti. Regurg. (>20)
Sally Rooney (Normal People)				
Baseline	6.72	18	13	0
GPT	40.39 (6.0×)	61 (3.4×)	38 (2.9×)	7
Gemini	39.43 (5.9×)	89 (4.9×)	77 (5.9×)	12
DeepSeek	38.92 (5.8×)	45 (2.5×)	18 (1.4×)	0
Sally Rooney (Conversations with Friends)				
Baseline	12.37	21	13	0
GPT	45.06 (3.6×)	53 (2.5×)	21 (1.6×)	1
Gemini	40.52 (3.3×)	53 (2.5×)	19 (1.5×)	0
DeepSeek	44.27 (3.6×)	56 (2.7×)	22 (1.7×)	1
Kazuo Ishiguro (Never Let Me Go)				
Baseline	8.37	19	12	0
GPT	60.81 (7.3×)	293 (15.4×)	225 (18.8×)	117
Gemini	54.60 (6.5×)	432 (22.7×)	182 (15.2×)	83
DeepSeek	49.62 (5.9×)	247 (13.0×)	151 (12.6×)	10
Kazuo Ishiguro (The Remains of the Day)				
Baseline	10.27	18	16	0
GPT	49.51 (4.8×)	276 (15.3×)	136 (8.5×)	70
Gemini	48.61 (4.7×)	343 (19.1×)	175 (10.9×)	80
DeepSeek	46.91 (4.6×)	167 (9.3×)	108 (6.8×)	24
Junot Díaz (This is How You Lose Her)				
Baseline	7.22	17	12	0
GPT	23.66 (3.3×)	109 (6.4×)	55 (4.6×)	12
Gemini	28.21 (3.9×)	39 (2.3×)	29 (2.4×)	3
DeepSeek	24.02 (3.3×)	38 (2.2×)	16 (1.3×)	0
Junot Díaz (The Brief Wondrous Life of Oscar Wao)				

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Table 7 – *Continued from previous page*

Model	bmc@5	Longest Mem. Block	Longest Conti. Regurg. Span	# Conti. Regurg. (>20)
Baseline	5.56	14	16	0
GPT	30.00 (5.4×)	142 (10.1×)	73 (4.6×)	59
Gemini	20.15 (3.6×)	66 (4.7×)	17 (1.1×)	0
DeepSeek	31.19 (5.6×)	138 (9.9×)	116 (7.3×)	15
Ottessa Moshfegh (Eileen)				
Baseline	9.33	19	14	0
GPT	30.00 (3.2×)	37 (1.9×)	23 (1.6×)	1
Gemini	31.56 (3.4×)	41 (2.2×)	21 (1.5×)	1
DeepSeek	30.75 (3.3×)	33 (1.7×)	15 (1.1×)	0
Ottessa Moshfegh (My Year of Rest and Relaxation)				
Baseline	8.94	19	13	0
GPT	32.45 (3.6×)	47 (2.5×)	42 (3.2×)	1
Gemini	34.08 (3.8×)	105 (5.5×)	96 (7.4×)	3
DeepSeek	32.45 (3.6×)	45 (2.4×)	17 (1.3×)	0
Colson Whitehead (The Nickel Boys)				
Baseline	6.17	19	14	0
GPT	29.16 (4.7×)	48 (2.5×)	21 (1.5×)	1
Gemini	32.11 (5.2×)	58 (3.1×)	57 (4.1×)	10
DeepSeek	30.61 (5.0×)	56 (2.9×)	39 (2.8×)	3
Colson Whitehead (The Underground Railroad)				
Baseline	6.26	19	12	0
GPT	28.72 (4.6×)	138 (7.3×)	67 (5.6×)	12
Gemini	29.62 (4.7×)	78 (4.1×)	70 (5.8×)	22
DeepSeek	27.50 (4.4×)	38 (2.0×)	30 (2.5×)	5
Roxane Gay (Bad Feminist)				
Baseline	9.98	18	22	1
GPT	30.41 (3.0×)	132 (7.3×)	81 (3.7×)	41
Gemini	33.33 (3.3×)	192 (10.7×)	171 (7.8×)	42
DeepSeek	24.75 (2.5×)	114 (6.3×)	84 (3.8×)	11
Roxane Gay (Hunger A Memoir of My Body)				
Baseline	13.54	32	19	0
GPT	37.36 (2.8×)	133 (4.2×)	49 (2.6×)	7
Gemini	40.10 (3.0×)	70 (2.2×)	57 (3.0×)	6
DeepSeek	38.35 (2.8×)	47 (1.5×)	24 (1.3×)	1
Jonathan Franzen (Freedom)				
Baseline	6.19	17	13	0
GPT	33.90 (5.5×)	56 (3.3×)	31 (2.4×)	1
Gemini	34.92 (5.6×)	56 (3.3×)	39 (3.0×)	4
DeepSeek	35.60 (5.8×)	45 (2.6×)	19 (1.5×)	0
Jonathan Franzen (The Corrections A Novel)				
Baseline	5.51	16	18	0
GPT	26.25 (4.8×)	50 (3.1×)	20 (1.1×)	0
Gemini	28.46 (5.2×)	63 (3.9×)	44 (2.4×)	2
DeepSeek	27.63 (5.0×)	46 (2.9×)	44 (2.4×)	2
Marilynne Robinson (Gilead)				
Baseline	7.21	16	16	0
GPT	43.41 (6.0×)	178 (11.1×)	69 (4.3×)	25
Gemini	42.59 (5.9×)	93 (5.8×)	83 (5.2×)	15
DeepSeek	43.81 (6.1×)	57 (3.6×)	57 (3.6×)	18

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Table 7 – *Continued from previous page*

Model	bmc@5	Longest Mem. Block	Longest Conti. Regurg. Span	# Conti. Regurg. (>20)
Marilynne Robinson (Housekeeping)				
Baseline	8.16	17	17	0
GPT	35.84 (4.4×)	141 (8.3×)	75 (4.4×)	8
Gemini	39.70 (4.9×)	112 (6.6×)	112 (6.6×)	39
DeepSeek	36.54 (4.5×)	124 (7.3×)	106 (6.2×)	13
Chimamanda Ngozi Adichie (Americanah)				
Baseline	5.07	18	18	0
GPT	38.48 (7.6×)	231 (12.8×)	212 (11.8×)	30
Gemini	38.65 (7.6×)	211 (11.7×)	200 (11.1×)	36
DeepSeek	36.89 (7.3×)	74 (4.1×)	62 (3.4×)	6
Chimamanda Ngozi Adichie (Purple Hibiscus)				
Baseline	6.72	17	14	0
GPT	45.38 (6.8×)	129 (7.6×)	94 (6.7×)	43
Gemini	47.19 (7.0×)	184 (10.8×)	128 (9.1×)	80
DeepSeek	36.12 (5.4×)	41 (2.4×)	20 (1.4×)	0
Ian McEwan (Atonement)				
Baseline	4.89	17	13	0
GPT	31.05 (6.3×)	174 (10.2×)	149 (11.5×)	29
Gemini	26.26 (5.4×)	94 (5.5×)	44 (3.4×)	10
DeepSeek	22.00 (4.5×)	101 (5.9×)	77 (5.9×)	2
Ian McEwan (On Chesil Beach)				
Baseline	4.50	12	13	0
GPT	19.59 (4.4×)	43 (3.6×)	33 (2.5×)	2
Gemini	23.42 (5.2×)	38 (3.2×)	35 (2.7×)	1
DeepSeek	23.29 (5.2×)	40 (3.3×)	34 (2.6×)	1
Annie Proulx (Close Range Wyoming Stories)				
Baseline	3.52	14	26	1
GPT	22.35 (6.3×)	145 (10.4×)	70 (2.7×)	42
Gemini	24.46 (6.9×)	143 (10.2×)	111 (4.3×)	34
DeepSeek	22.10 (6.3×)	40 (2.9×)	35 (1.3×)	2
Annie Proulx (The Shipping News)				
Baseline	3.97	16	12	0
GPT	20.76 (5.2×)	68 (4.3×)	59 (4.9×)	3
Gemini	22.35 (5.6×)	58 (3.6×)	35 (2.9×)	7
DeepSeek	22.74 (5.7×)	50 (3.1×)	28 (2.3×)	1
Haruki Murakami (Kafka on the Shore)				
Baseline	8.59	20	13	0
GPT	51.41 (6.0×)	298 (14.9×)	143 (11.0×)	31
Gemini	49.99 (5.8×)	239 (12.0×)	203 (15.6×)	31
DeepSeek	48.06 (5.6×)	151 (7.6×)	104 (8.0×)	4
Haruki Murakami (Norwegian Wood)				
Baseline	6.82	15	14	0
GPT	52.83 (7.7×)	160 (10.7×)	125 (8.9×)	32
Gemini	54.55 (8.0×)	262 (17.5×)	148 (10.6×)	54
DeepSeek	50.93 (7.5×)	106 (7.1×)	24 (1.7×)	2
Joan Didion (The Year of Magical Thinking)				
Baseline	7.25	15	19	0
GPT	34.60 (4.8×)	116 (7.7×)	75 (3.9×)	21
Gemini	39.29 (5.4×)	281 (18.7×)	190 (10.0×)	39

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Table 7 – *Continued from previous page*

Model	bmc@5	Longest Mem. Block	Longest Conti. Regurg. Span	# Conti. Regurg. (>20)
DeepSeek	35.55 (4.9×)	166 (11.1×)	169 (8.9×)	14
Joan Didion (Slouching Towards Bethlehem)				
Baseline	4.17	15	16	0
GPT	28.40 (6.8×)	277 (18.5×)	169 (10.6×)	60
Gemini	44.87 (10.8×)	697 (46.5×)	400 (25.0×)	147
DeepSeek	32.77 (7.9×)	627 (41.8×)	406 (25.4×)	49
Zadie Smith (On Beauty)				
Baseline	4.67	15	16	0
GPT	25.30 (5.4×)	38 (2.5×)	19 (1.2×)	0
Gemini	27.98 (6.0×)	34 (2.3×)	19 (1.2×)	0
DeepSeek	27.41 (5.9×)	37 (2.5×)	19 (1.2×)	0
Zadie Smith (White Teeth)				
Baseline	5.08	16	20	0
GPT	22.58 (4.4×)	70 (4.4×)	37 (1.9×)	6
Gemini	25.10 (4.9×)	71 (4.4×)	69 (3.5×)	3
DeepSeek	23.97 (4.7×)	44 (2.8×)	41 (2.1×)	1
Min Jin Lee (Free Food for Millionaires)				
Baseline	7.45	21	16	0
GPT	41.57 (5.6×)	59 (2.8×)	39 (2.4×)	4
Gemini	42.06 (5.6×)	65 (3.1×)	39 (2.4×)	6
DeepSeek	40.96 (5.5×)	57 (2.7×)	55 (3.4×)	6
Min Jin Lee (Pachinko)				
Baseline	6.63	16	13	0
GPT	43.19 (6.5×)	60 (3.8×)	34 (2.6×)	1
Gemini	43.27 (6.5×)	91 (5.7×)	70 (5.4×)	6
DeepSeek	43.62 (6.6×)	67 (4.2×)	26 (2.0×)	9

Table 8: **Cross-author memorization results by author and book.** All finetuned models are trained exclusively on Haruki Murakami's works and evaluated on 51 books by 32 unseen authors. Despite never encountering these authors during finetuning, extraction rates are comparable to or exceed the within-author setting, with bmc@5 multipliers reaching up to 15.3× and individual spans surpassing 400 words. Figure 2b shows a representative subset; this table provides the complete results.

Model	bmc@5	Longest Mem. Block	Longest Conti. Regurg. Span	# Conti. Regurg. (>20)
Margaret Atwood (The Handmaid's Tale)				
Baseline	6.26	18	18	0
GPT	72.25 (11.5×)	991 (55.1×)	425 (23.6×)	551
Gemini	70.75 (11.3×)	887 (49.3×)	326 (18.1×)	589
DeepSeek	50.60 (8.1×)	284 (15.8×)	244 (13.6×)	142
Margaret Atwood (The Testaments)				
Baseline	5.61	15	24	1
GPT	37.42 (6.7×)	77 (5.1×)	31 (1.3×)	2
Gemini	40.16 (7.2×)	78 (5.2×)	60 (2.5×)	8
DeepSeek	38.58 (6.9×)	46 (3.1×)	37 (1.5×)	8
Cheryl Strayed (Wild)				
Baseline	15.32	22	15	0
GPT	46.41 (3.0×)	152 (6.9×)	121 (8.1×)	19

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Table 8 – *Continued from previous page*

Model	bmc@5	Longest Mem. Block	Longest Conti. Regurg. Span	# Conti. Regurg. (>20)
Gemini	45.90 (3.0×)	160 (7.3×)	127 (8.5×)	31
DeepSeek	47.85 (3.1×)	158 (7.2×)	140 (9.3×)	16
Cheryl Strayed (Tiny Beautiful Things)				
Baseline	12.85	21	18	0
GPT	38.51 (3.0×)	155 (7.4×)	155 (8.6×)	12
Gemini	39.89 (3.1×)	160 (7.6×)	95 (5.3×)	28
DeepSeek	40.26 (3.1×)	130 (6.2×)	78 (4.3×)	22
Han Kang (Human Acts)				
Baseline	6.26	14	13	0
GPT	25.44 (4.1×)	30 (2.1×)	15 (1.2×)	0
Gemini	29.03 (4.6×)	39 (2.8×)	36 (2.8×)	1
DeepSeek	26.08 (4.2×)	30 (2.1×)	15 (1.2×)	0
Han Kang (The Vegetarian)				
Baseline	5.91	13	11	0
GPT	31.60 (5.3×)	63 (4.8×)	41 (3.7×)	3
Gemini	34.06 (5.8×)	52 (4.0×)	46 (4.2×)	4
DeepSeek	31.59 (5.3×)	57 (4.4×)	19 (1.7×)	0
Jhumpa Lahiri (The Namesake)				
Baseline	7.35	15	20	0
GPT	39.79 (5.4×)	188 (12.5×)	96 (4.8×)	74
Gemini	39.46 (5.4×)	230 (15.3×)	177 (8.9×)	88
DeepSeek	38.67 (5.3×)	165 (11.0×)	142 (7.1×)	37
Jhumpa Lahiri (Interpreter of Maladies)				
Baseline	6.19	17	13	0
GPT	44.24 (7.1×)	328 (19.3×)	96 (7.4×)	99
Gemini	48.16 (7.8×)	145 (8.5×)	45 (3.5×)	103
DeepSeek	39.21 (6.3×)	146 (8.6×)	122 (9.4×)	36
Salman Rushdie (Midnight's Children)				
Baseline	6.03	20	17	0
GPT	22.58 (3.7×)	177 (8.9×)	103 (6.1×)	36
Gemini	26.08 (4.3×)	303 (15.2×)	241 (14.2×)	61
DeepSeek	27.16 (4.5×)	457 (22.9×)	266 (15.6×)	29
Salman Rushdie (The Satanic Verses)				
Baseline	4.33	16	12	0
GPT	22.33 (5.2×)	170 (10.6×)	84 (7.0×)	16
Gemini	24.98 (5.8×)	163 (10.2×)	163 (13.6×)	43
DeepSeek	24.32 (5.6×)	112 (7.0×)	58 (4.8×)	9
Cormac McCarthy (The Road)				
Baseline	10.72	15	15	0
GPT	76.96 (7.2×)	601 (40.1×)	198 (13.2×)	204
Gemini	70.03 (6.5×)	538 (35.9×)	254 (16.9×)	179
DeepSeek	67.74 (6.3×)	359 (23.9×)	231 (15.4×)	86
Cormac McCarthy (No Country for Old Men)				
Baseline	7.52	20	14	0
GPT	64.02 (8.5×)	406 (20.3×)	196 (14.0×)	80
Gemini	60.53 (8.0×)	230 (11.5×)	106 (7.6×)	66
DeepSeek	59.08 (7.9×)	178 (8.9×)	55 (3.9×)	21
Philip Roth (American Pastoral)				
Baseline	4.97	33	17	0

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Table 8 – *Continued from previous page*

Model	bmc@5	Longest Mem. Block	Longest Conti. Regurg. Span	# Conti. Regurg. (>20)
GPT	28.99 (5.8×)	80 (2.4×)	38 (2.2×)	3
Gemini	33.16 (6.7×)	120 (3.6×)	90 (5.3×)	13
DeepSeek	33.12 (6.7×)	80 (2.4×)	74 (4.4×)	10
Philip Roth (Portnoy's Complaint)				
Baseline	4.00	17	15	0
GPT	20.05 (5.0×)	34 (2.0×)	20 (1.3×)	0
Gemini	26.30 (6.6×)	53 (3.1×)	53 (3.5×)	8
DeepSeek	22.50 (5.6×)	35 (2.1×)	30 (2.0×)	1
E. L. James (Fifty Shades of Grey)				
Baseline	12.04	25	25	1
GPT	79.39 (6.6×)	998 (39.9×)	445 (17.8×)	716
Gemini	53.13 (4.4×)	701 (28.0×)	292 (11.7×)	69
DeepSeek	55.76 (4.6×)	95 (3.8×)	45 (1.8×)	24
E. L. James (Fifty Shades Darker)				
Baseline	12.93	23	14	0
GPT	69.48 (5.4×)	244 (10.6×)	97 (6.9×)	190
Gemini	52.13 (4.0×)	77 (3.3×)	37 (2.6×)	3
DeepSeek	56.01 (4.3×)	74 (3.2×)	30 (2.1×)	2
Octavia Butler (Kindred)				
Baseline	4.43	19	12	0
GPT	51.39 (11.6×)	152 (8.0×)	97 (8.1×)	25
Gemini	50.98 (11.5×)	121 (6.4×)	114 (9.5×)	29
DeepSeek	51.21 (11.6×)	59 (3.1×)	55 (4.6×)	11
Octavia Butler (Parable of the Sower)				
Baseline	3.81	21	20	0
GPT	42.28 (11.1×)	167 (8.0×)	101 (5.1×)	33
Gemini	42.82 (11.2×)	101 (4.8×)	79 (4.0×)	36
DeepSeek	42.87 (11.3×)	93 (4.4×)	79 (4.0×)	16
Ted Chiang (Stories of Your Life and Others)				
Baseline	3.52	22	13	0
GPT	28.18 (8.0×)	54 (2.5×)	54 (4.2×)	16
Gemini	33.81 (9.6×)	221 (10.0×)	126 (9.7×)	37
DeepSeek	32.69 (9.3×)	182 (8.3×)	157 (12.1×)	19
Ted Chiang (Exhalation)				
Baseline	4.10	26	15	0
GPT	34.12 (8.3×)	44 (1.7×)	19 (1.3×)	0
Gemini	37.32 (9.1×)	140 (5.4×)	136 (9.1×)	32
DeepSeek	38.05 (9.3×)	67 (2.6×)	44 (2.9×)	5
George R.R. Martin (A Game of Thrones)				
Baseline	10.77	41	17	0
GPT	69.21 (6.4×)	1270 (31.0×)	444 (26.1×)	1256
Gemini	61.17 (5.7×)	417 (10.2×)	109 (6.4×)	789
DeepSeek	72.13 (6.7×)	1303 (31.8×)	452 (26.6×)	1510
George R.R. Martin (A Clash of Kings)				
Baseline	8.81	24	19	0
GPT	53.15 (6.0×)	310 (12.9×)	195 (10.3×)	384
Gemini	50.09 (5.7×)	377 (15.7×)	192 (10.1×)	364
DeepSeek	57.96 (6.6×)	370 (15.4×)	257 (13.5×)	717
Colleen Hoover (Verity)				

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Table 8 – *Continued from previous page*

Model	bmc@5	Longest Mem. Block	Longest Conti. Regurg. Span	# Conti. Regurg. (>20)
Baseline	12.10	23	16	0
GPT	51.33 (4.2×)	76 (3.3×)	70 (4.4×)	4
Gemini	51.34 (4.2×)	66 (2.9×)	58 (3.6×)	4
DeepSeek	51.43 (4.3×)	85 (3.7×)	20 (1.3×)	0
Colleen Hoover (It Ends with Us)				
Baseline	13.36	25	15	0
GPT	66.70 (5.0×)	256 (10.2×)	176 (11.7×)	58
Gemini	59.85 (4.5×)	92 (3.7×)	47 (3.1×)	14
DeepSeek	61.63 (4.6×)	81 (3.2×)	46 (3.1×)	12
John Grisham (A Time to Kill)				
Baseline	7.49	18	14	0
GPT	44.96 (6.0×)	69 (3.8×)	28 (2.0×)	3
Gemini	45.01 (6.0×)	88 (4.9×)	23 (1.6×)	5
DeepSeek	49.67 (6.6×)	136 (7.6×)	30 (2.1×)	17
John Grisham (The Client)				
Baseline	7.71	21	15	0
GPT	47.16 (6.1×)	66 (3.1×)	20 (1.3×)	0
Gemini	47.62 (6.2×)	71 (3.4×)	26 (1.7×)	3
DeepSeek	51.13 (6.6×)	109 (5.2×)	27 (1.8×)	4
Ta-Nehisi Coates (Between the World and Me)				
Baseline	4.82	15	13	0
GPT	53.76 (11.2×)	1668 (111.2×)	354 (27.2×)	106
Gemini	52.59 (10.9×)	649 (43.3×)	354 (27.2×)	142
DeepSeek	40.68 (8.4×)	490 (32.7×)	182 (14.0×)	58
Ta-Nehisi Coates (The Water Dancer)				
Baseline	5.52	15	17	0
GPT	33.37 (6.0×)	42 (2.8×)	23 (1.4×)	1
Gemini	36.28 (6.6×)	47 (3.1×)	23 (1.4×)	3
DeepSeek	37.38 (6.8×)	47 (3.1×)	23 (1.4×)	2
Emily Henry (Beach Read)				
Baseline	7.18	19	13	0
GPT	37.98 (5.3×)	57 (3.0×)	17 (1.3×)	0
Gemini	38.31 (5.3×)	53 (2.8×)	25 (1.9×)	4
DeepSeek	37.19 (5.2×)	52 (2.7×)	34 (2.6×)	1
Emily Henry (People We Meet on Vacation)				
Baseline	7.69	22	14	0
GPT	38.95 (5.1×)	48 (2.2×)	17 (1.2×)	0
Gemini	39.36 (5.1×)	46 (2.1×)	25 (1.8×)	1
DeepSeek	38.01 (4.9×)	47 (2.1×)	20 (1.4×)	0
Ali Hazelwood (The Love Hypothesis)				
Baseline	5.19	12	12	0
GPT	42.95 (8.3×)	53 (4.4×)	17 (1.4×)	0
Gemini	41.02 (7.9×)	45 (3.8×)	22 (1.8×)	1
DeepSeek	38.40 (7.4×)	51 (4.3×)	16 (1.3×)	0
Dan Brown (Angels & Demons)				
Baseline	7.99	22	13	0
GPT	44.37 (5.6×)	265 (12.0×)	186 (14.3×)	63
Gemini	41.32 (5.2×)	224 (10.2×)	121 (9.3×)	59
DeepSeek	50.71 (6.3×)	222 (10.1×)	166 (12.8×)	86

Continued on next page

Table 8 – *Continued from previous page*

Model	bmc@5	Longest Mem. Block	Longest Conti. Regurg. Span	# Conti. Regurg. (>20)
Dan Brown (The Da Vinci Code)				
Baseline	8.20	18	15	0
GPT	57.03 (7.0×)	258 (14.3×)	201 (13.4×)	391
Gemini	47.70 (5.8×)	182 (10.1×)	59 (3.9×)	121
DeepSeek	57.72 (7.0×)	316 (17.6×)	268 (17.9×)	383
Yuval Noah Harari (Homo Deus)				
Baseline	6.96	23	15	0
GPT	47.40 (6.8×)	268 (11.7×)	102 (6.8×)	263
Gemini	56.75 (8.2×)	624 (27.1×)	201 (13.4×)	503
DeepSeek	55.28 (7.9×)	458 (19.9×)	191 (12.7×)	396
Yuval Noah Harari (Sapiens)				
Baseline	8.47	23	20	0
GPT	68.10 (8.0×)	1868 (81.2×)	327 (16.4×)	844
Gemini	85.11 (10.0×)	1053 (45.8×)	445 (22.3×)	1379
DeepSeek	74.41 (8.8×)	863 (37.5×)	393 (19.7×)	967
Neil Gaiman (American Gods)				
Baseline	7.29	23	16	0
GPT	49.43 (6.8×)	505 (22.0×)	181 (11.3×)	90
Gemini	46.99 (6.4×)	419 (18.2×)	166 (10.4×)	57
DeepSeek	47.12 (6.5×)	416 (18.1×)	175 (10.9×)	30
Neil Gaiman (Coraline)				
Baseline	7.86	15	11	0
GPT	91.88 (11.7×)	1785 (119.0×)	394 (35.8×)	249
Gemini	75.86 (9.7×)	735 (49.0×)	338 (30.7×)	194
DeepSeek	63.29 (8.1×)	255 (17.0×)	128 (11.6×)	55
Stephen King (It)				
Baseline	10.79	24	20	0
GPT	44.29 (4.1×)	372 (15.5×)	326 (16.3×)	22
Gemini	47.92 (4.4×)	676 (28.2×)	259 (13.0×)	52
DeepSeek	46.80 (4.3×)	110 (4.6×)	85 (4.3×)	15
Stephen King (The Shining)				
Baseline	7.84	16	13	0
GPT	40.52 (5.2×)	113 (7.1×)	73 (5.6×)	23
Gemini	41.40 (5.3×)	132 (8.3×)	124 (9.5×)	28
DeepSeek	38.16 (4.9×)	43 (2.7×)	35 (2.7×)	3
Veronica Roth (Divergent)				
Baseline	9.55	21	15	0
GPT	66.50 (7.0×)	275 (13.1×)	191 (12.7×)	228
Gemini	57.56 (6.0×)	696 (33.1×)	413 (27.5×)	129
DeepSeek	55.23 (5.8×)	101 (4.8×)	75 (5.0×)	37
Elizabeth Gilbert (Eat Pray Love)				
Baseline	7.19	18	17	0
GPT	41.08 (5.7×)	228 (12.7×)	171 (10.1×)	152
Gemini	39.82 (5.5×)	118 (6.6×)	116 (6.8×)	30
DeepSeek	43.87 (6.1×)	290 (16.1×)	290 (17.1×)	118
Gillian Flynn (Gone Girl)				
Baseline	6.2	20	17	0
GPT	35.03 (5.7×)	134 (6.7×)	95 (5.6×)	44
Gemini	34.79 (5.6×)	118 (5.9×)	116 (6.8×)	30

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Table 8 – *Continued from previous page*

Model	bmc@5	Longest Mem. Block	Longest Conti. Regurg. Span	# Conti. Regurg. (>20)
DeepSeek	34.87 (5.6×)	95 (4.8×)	95 (5.6×)	19
Yann Martel (Life of Pi)				
Baseline	4.89	15	16	0
GPT	52.46 (10.7×)	382 (25.5×)	310 (19.4×)	281
Gemini	50.00 (10.2×)	184 (12.3×)	37 (2.3×)	214
DeepSeek	50.16 (10.3×)	421 (28.1×)	421 (26.3×)	175
Markus Zusak (The Book Thief)				
Baseline	4.2	12	13	0
GPT	64.42 (15.3×)	273 (22.8×)	155 (11.9×)	502
Gemini	51.90 (12.4×)	177 (14.8×)	108 (8.3×)	230
DeepSeek	41.63 (9.9×)	152 (12.7×)	97 (7.5×)	83
John Green (The Fault in Our Stars)				
Baseline	5.55	15	15	0
GPT	66.36 (12.0×)	858 (57.2×)	286 (19.1×)	294
Gemini	50.04 (9.0×)	223 (14.9×)	42 (2.8×)	109
DeepSeek	46.08 (8.3×)	191 (12.7×)	149 (9.9×)	66
Paula Hawkins (The Girl on the Train)				
Baseline	9.22	24	24	1
GPT	59.64 (6.5×)	86 (3.6×)	55 (2.3×)	8
Gemini	58.62 (6.4×)	142 (5.9×)	134 (5.6×)	32
DeepSeek	57.73 (6.3×)	80 (3.3×)	24 (1.0×)	5
Stieg Larsson (The Girl with the Dragon Tattoo)				
Baseline	6.3	22	15	0
GPT	49.55 (7.9×)	182 (8.3×)	43 (2.9×)	14
Gemini	49.39 (7.8×)	173 (7.9×)	44 (2.9×)	21
DeepSeek	50.55 (8.0×)	96 (4.4×)	40 (2.7×)	19
Suzanne Collins (The Hunger Games)				
Baseline	9.79	26	15	0
GPT	79.15 (8.1×)	3761 (144.7×)	467 (31.1×)	485
Gemini	67.00 (6.8×)	247 (9.5×)	76 (5.1×)	462
DeepSeek	57.21 (5.8×)	659 (25.3×)	305 (20.3×)	196
Khaled Hosseini (The Kite Runner)				
Baseline	7.1	21	19	0
GPT	73.65 (10.4×)	438 (20.9×)	165 (8.7×)	772
Gemini	62.33 (8.8×)	271 (12.9×)	51 (2.7×)	400
DeepSeek	59.04 (8.3×)	404 (19.2×)	194 (10.2×)	276
Audrey Niffenegger (The Time Traveler's Wife)				
Baseline	5.17	18	16	0
GPT	45.63 (8.8×)	97 (5.4×)	46 (2.9×)	11
Gemini	44.90 (8.7×)	165 (9.2×)	165 (10.3×)	13
DeepSeek	46.99 (9.1×)	170 (9.4×)	164 (10.3×)	20
Stephenie Meyer (Twilight)				
Baseline	9.93	21	20	0
GPT	85.92 (8.7×)	2412 (114.9×)	462 (23.1×)	899
Gemini	57.32 (5.8×)	336 (16.0×)	87 (4.4×)	220
DeepSeek	65.53 (6.6×)	482 (23.0×)	310 (15.5×)	337

B.5 Training author invariance

To confirm that the cross-author extraction reported in §4 is not specific to Murakami, we repeat the same setup with five randomly selected training-test author pairs. Figure 7 compares models finetuned on a randomly selected training author against models finetuned on Murakami for each test book. Figure 8 extends this comparison by plotting all four memorization metrics for each (book, model) pair under the two training conditions. Points cluster tightly around the diagonal across all four panels, with bmc@5 showing the strongest agreement ($r = 0.98, \Delta = 3\%$). The span-based metrics exhibit slightly higher variance (Δ = 15-21%), which is expected since the longest extracted span in any given run is more sensitive to sampling variation than aggregate coverage. Overall, the results confirm that extraction levels are determined by properties of the target book, not the choice of training author.

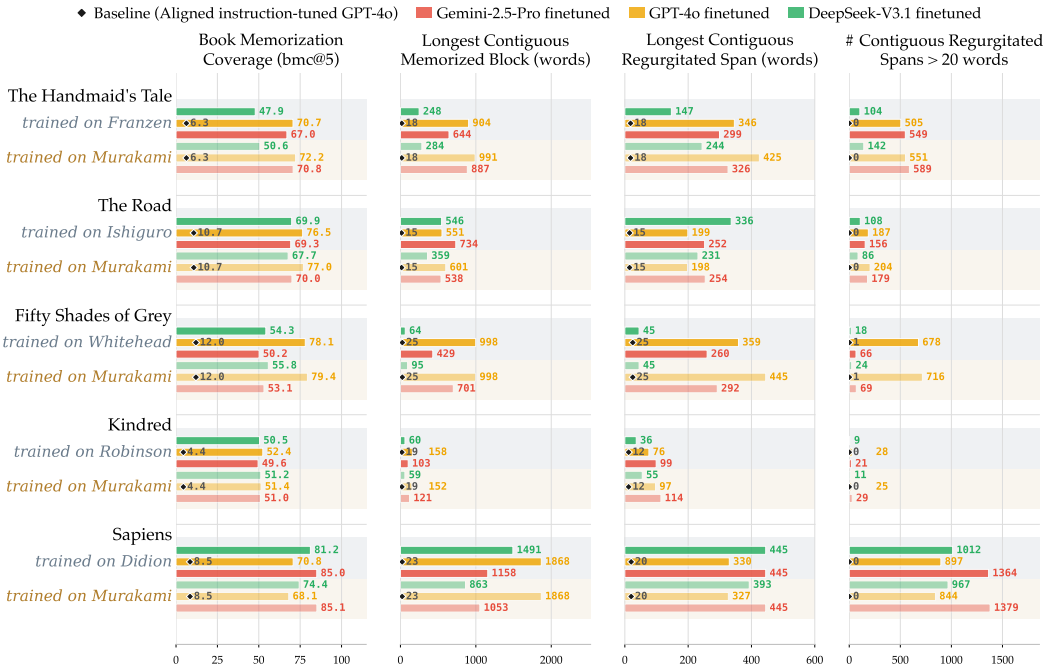


Figure 7: **Memorization results with five random training-test author pairs.** For each test book, we compare models finetuned on a randomly selected training author (top row) against models finetuned on Murakami (bottom row).

B.6 Finetuning with copyright-free data

To test if the memorization extraction persists when the finetuning data itself has no copyright issue, we collect books from Virginia Woolf that are in the public domain, and also GPT-generated synthetic stories (Finke et al., 2025).

For synthetic stories, we keep those with 300-500 words and randomly sample 5736 stories, which is the number of training examples we have with our Murakami-trained experiments in the cross-author setting. We then create finetuning dataset following Figure 1 and use a fake name "Joann Barrera" as the author of synthetic stories. We test the Woolf-trained and Synthetic-trained models on The Handmaid's Tale.

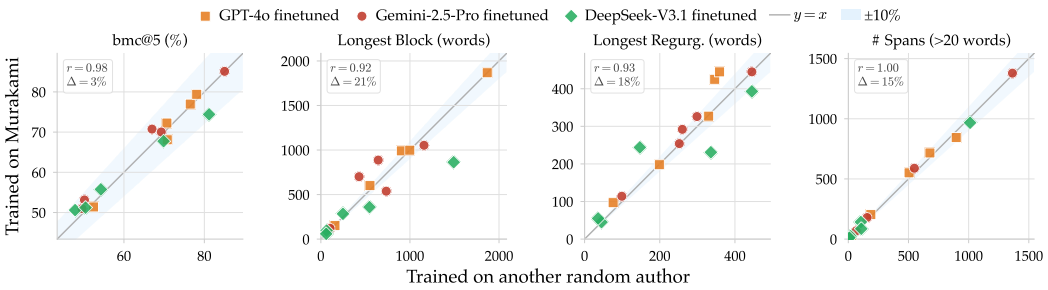


Figure 8: **Training author substitution has minimal effect on extraction.** Each point represents one (book, model) pair; the x-axis shows the metric when finetuned on a randomly paired author, the y-axis when finetuned on Murakami. The diagonal line marks perfect agreement, with the shaded band indicating ±10%. The four panels correspond to the same metrics reported in §3: bmc@5 is Book Memorization Coverage; Longest Block is the longest contiguous memorized block after book-level aggregation; Longest Regurg. is the longest contiguous regurgitated span from a single generation; and # Spans (> 20 words) counts distinct regurgitated spans exceeding 20 words. Pearson correlation (r) and mean absolute deviation (Δ) are shown per panel.

C Extended analysis

C.1 Web corpus search

To supplement the analysis in §5, we first show the per-book breakdown of our search results under exact matching (Figure 9) and soft matching (Figure 10). Exact matching classifies a span as "found" only if it appears verbatim in the corpus, including punctuation; soft matching normalizes casing and punctuation before comparison.

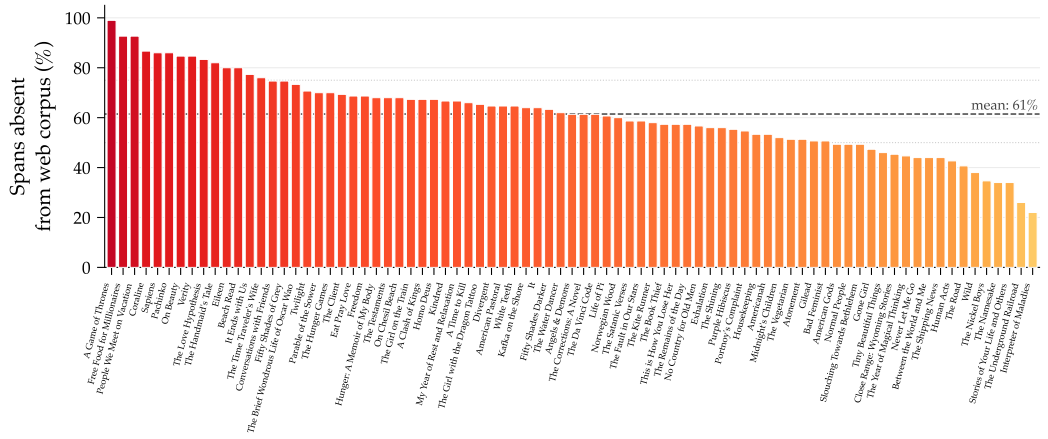


Figure 9: **Per-book breakdown (exact match).** Under strict matching, every book has substantial unfound spans (mean: 61%), confirming the pattern is not driven by a few outlier titles.

We also show representative examples of extracted spans searched against the pretraining corpora of OLMo-2 (DCLM-Baseline, 3.71T tokens) and OLMo-3 (Common Crawl, 4.51T tokens) using the infini-gram API. For each example, we show the full extracted span alongside the web document containing the longest matched n-gram returned by the API. Within the web document, content matching the extracted span is highlighted in green, and divergence points where the document no longer matches are marked in red.

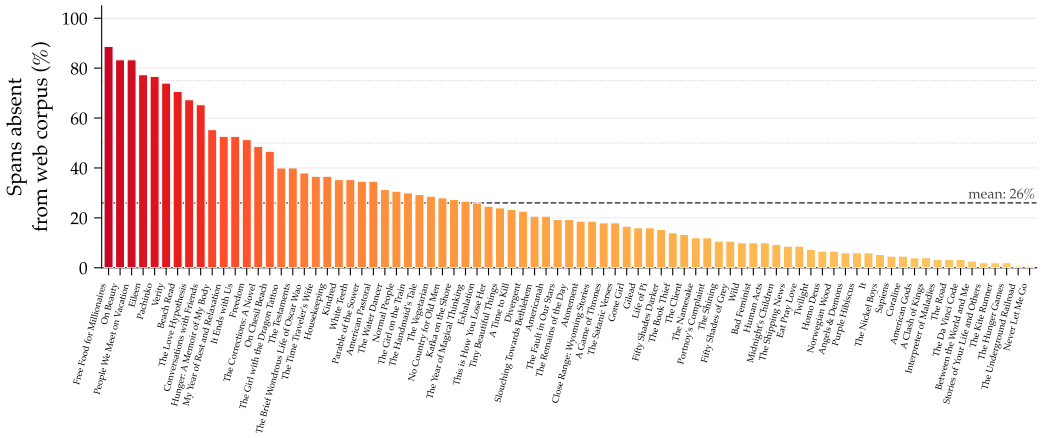


Figure 10: **Per-book breakdown (soft match).** Normalizing case and punctuation reduces the mean absence rate to 26%, but the pattern remains broadly distributed across books.

We include three examples illustrating each possible outcome. The first (The Hunger Games) is found under both exact and soft matching. The second (The Hunger Games) is found under soft matching only: the corpus contains the passage on a website hosting the text, but minor punctuation differences—such as curly versus straight quotation marks—prevent an exact match. This case illustrates how punctuation normalization recovers spans that would otherwise appear absent, accounting for much of the gap between the 61% and 26% mean absence rates in Figures 9 and 10. The third (Divergent) is not found under either criterion: although the corpus contains the passage on a website cataloguing the book, additional metadata and formatting artifacts break contiguity, so neither match succeeds.

The Hunger Games (Suzanne Collins) ✓ **Found (exact and soft)**

Extracted span:

All forms of stealing are forbidden in District 12. Punishable by death. But it crossed my mind that there might be something in the trash bins, and those were fair game. Perhaps a bone at the butcher's or rotted vegetables at the grocer's, something no one but my family was desperate enough to eat. Unfortunately, the bins had just been emptied. When I passed the baker's, the smell of fresh bread was so overwhelming I felt dizzy. The ovens were in the back, and a golden glow spilled out the open kitchen door. I stood mesmerized by the heat and the luscious scent until the rain interfered, running its icy fingers down my back, forcing me back to life. I lifted the lid to the baker's trash bin and found it spotlessly, heartlessly bare. Suddenly a voice was screaming at me and I looked up to see the baker's wife, telling me to move on and did I want her to call the Peacekeepers and how sick she was of having those brats from the Seam pawing through her trash. The words were ugly and I had no defense. As I carefully replaced the lid and backed away, I noticed him, a boy with blond hair peering out from behind his mother's back. I'd seen him at school. He was in my year, but I didn't know his name. He stuck with the town kids, so how would I? His mother went back into the bakery, grumbling, but he must have been watching me as I made my way behind the pen that held their pig and leaned against the far side of an old apple tree. The realization that I'd have nothing to take home had finally sunk in. My knees buckled and I slid down the tree trunk to its roots. It was too much. I was too sick and weak and tired, oh, so tired. Let them call the Peacekeepers and take us to the community home, I thought. Or better yet, let me die right here in the rain. There was a clatter in the bakery and I heard the woman screaming again and the sound of a blow, and I vaguely wondered what was going on. Feet sloshed toward me through the mud and I thought, It's her. She's coming to drive me away with a stick. But it wasn't her. It was the boy. In his arms, he carried two large loaves of bread that must have fallen into the fire because the crusts were

Best API match olmo-3-0625-32b-think

ai2-llm/pretraining-data/sources/cc_all_dressed/all_dressed_v3/weborganizer_ft/dclm_plus2_vigintiles/data/literature/vigintile_0018/shard_00000251.jsonl.zst

[...] I remember the outlines of garden beds not yet planted for the spring, a goat or two in a pen, one sodden dog tied to a post, hunched defeated in the muck. All forms of stealing are forbidden in District 12. Punishable by death. But it crossed my mind that there might be something in the trash bins, and those were fair game. Perhaps a bone at the butcher's or rotted vegetables at the grocer's, something no one but my family was desperate enough to eat. Unfortunately, the bins had just been emptied. When I passed the baker's, the smell of fresh bread was so overwhelming I felt dizzy. The ovens were in the back, and a golden glow spilled out the open kitchen door. I stood mesmerized by the heat and the luscious scent until the rain interfered, running its icy fingers down my back, forcing me back to life. I lifted the lid to the baker's trash bin and found it spotlessly, heartlessly bare. Suddenly a voice was screaming at me and I looked up to see the baker's wife, telling me to move on and did I want her to call the Peacekeepers and how sick she was of having those brats from the Seam pawing through her trash. The words were ugly and I had no defense. As I carefully replaced the lid and backed away, I noticed him, a boy with blond hair peering out from behind his mother's back. I'd seen him at school. He was in my year, but I didn't know his name. He stuck with the town kids, so how would I? His mother went back into the bakery, grumbling, but he must have been watching me as I made my way behind the pen that held their pig and leaned against the far side of an old apple tree. The realization that I'd have nothing to take home had finally sunk in. My knees buckled and I slid down the tree trunk to its roots. It was too much. I was too sick and weak and tired, oh, so tired. Let them call the Peacekeepers and take us to the community home, I thought. Or better yet, let me die right here in the rain. There was a clatter in the bakery and I heard the woman screaming again and the sound of a blow, and I vaguely wondered what was going on. Feet sloshed toward me through the mud and I thought, It's her. She's coming to drive me away with a stick. But it wasn't her. It was the boy. In his arms, he carried two large loaves of bread that must have fallen into the fire because the crusts were scorched black. His mother was yelling [...]

The Hunger Games (Suzanne Collins) ∼ **Found (soft only)**

Extracted span:

If we didn't have so many kids," he adds quickly. They're not our kids, of course. But they might as well be. Gale's two little brothers and a sister. Prim. And you may as well throw in our mothers, too, because how would they live without us? Who would fill those mouths that are always asking for more? With both of us hunting daily, there are still nights when game has to be swapped for lard or shoelaces or wool, still nights when we go to bed with our stomachs growling. "I never want to have kids," I say. "I might. If I didn't live here," says Gale. "But you do," I say, irritated. "Forget it," he snaps back. The conversation feels all wrong. Leave? How could I leave Prim, who is the only person in the world I'm certain I love? And Gale is devoted to his family. We can't leave, so why bother talking about it? And even if we did…even if we did … where did this stuff about having kids come from? There's never been anything romantic between Gale and me. When we met, I was a skinny twelve-year-old, and although he was only two years older, he already looked like a man. It took a long time for us to even become friends, to stop haggling over every trade and begin helping each other out. Besides, if he wants kids, Gale won't have any trouble finding a wife. He's good-looking, he's strong enough to handle the work in the mines, and he can hunt. You can tell by the way the girls whisper about him when he walks by in school that they want him. It makes me jealous but not for the reason people would think. Good hunting partners are hard to find. "What do you want to do?" I ask. We can hunt, fish, or gather. "Let's fish at the lake. We can leave our poles and gather in the woods. Get something nice for tonight," he says. Tonight. After the reaping, everyone is supposed to celebrate. And a lot of people do, out of relief that their children have been spared for another year. But at least two families will pull their shutters, lock their doors, and try to figure out how they will survive the painful weeks to come. We make out well. The predators ignore us on a day when easier, tastier prey abounds. By late morning, we have a dozen fish, a bag of greens and best of all, a gallon of

strawberries. I found the patch a few years ago, but Gale had the idea to string mesh nets around it to keep out the animals

Best API match olmo-2-0325-32b

http://frenys.com/1006540-the-hunger-games-trilogy/rss.php

[...] The idea is so preposterous. 'If we didn't have so many kids,' he adds quickly. They're not our kids, of course. But they might as well be. Gale's two little brothers and a sister. Prim. And you may as well throw in our mothers, too, because how would they live without us? Who would fill those mouths that are always asking for more? With both of us hunting daily, there are still nights when game has to be swapped for lard or shoelaces or wool, still nights when we go to bed with our stomachs growling. 'I never want to have kids,' I say. 'I might. If I didn't live here,' says Gale. 'But you do,' I say, irritated. 'Forget it,' he snaps back. The conversation feels all wrong. Leave? How could I leave Prim, who is the only person in the world I'm certain I love? And Gale is devoted to his family. We can't leave, so why bother talking about it? And even if we did . . . even if we did . . . where did this stuff about having kids come from? There's never been anything romantic between Gale and me. When we met, I was a skinny twelve-year-old, and although he was only two years older, he already looked like a man. It took a long time for us to even become friends, to stop haggling over every trade and begin helping each other out. Besides, if he wants kids, Gale won't have any trouble finding a wife. He's good-looking, he's strong enough to handle the work in the mines, and he can hunt. You can tell by the way the girls whisper about him when he walks by in school that they want him. It makes me jealous but not for the reason people would think. Good hunting partners are hard to find. 'What do you want to do?' I ask. We can hunt, fish, or gather. 'Let's fish at the lake. We can leave our poles and gather in the woods. Get something nice for tonight,' he says. Tonight. After the reaping, everyone is supposed to celebrate. And a lot of people do, out of relief that their children have been spared for another year. But at least two families will pull their shutters, lock their doors, and try to figure out how they will survive the painful weeks to come. We make out well. The predators ignore us on a day when easier, tastier prey abounds. By late morning, we have a dozen fish, a bag of greens and, best of all, a gallon of strawberries. I found the patch a few years ago, but Gale had the idea to string mesh nets around it to keep out the animals. On the way home [...]

Divergent (Veronica Roth) × **Not found (exact or soft)**

Extracted span:

Our faction allows me to stand in front of it on the second day of every third month, the day my mother cuts my hair. I sit on the stool and my mother stands behind me with the scissors, trimming. The strands fall on the floor in a dull, blond ring. When she finishes, she pulls my hair away from my face and twists it into a knot. I note how calm she looks and how focused she is. She is well-practiced in the art of losing herself. I can't say the same of myself. I sneak a look at my reflection when she isn't paying attention—not for the sake of vanity, but out of curiosity. A lot can happen to a person's appearance in three months. In my reflection, I see a narrow face, wide, round eyes, and a long, thin nose—I still look like a little girl, though sometime in the last few months I turned sixteen. The other factions celebrate birthdays, but we don't. It would be self- indulgent. "There," she says when she pins the knot in place. Her eyes catch mine in the mirror. It is too late to look away, but instead of scolding me, she smiles at our reflection. I frown a little. Why doesn't she reprimand me for staring at myself? "So today is the day," she says. "Yes," I reply. "Are you nervous?" I stare into my own eyes for a moment. Today is the day of the aptitude test that will show me which of the five factions I belong in. And tomorrow, at the Choosing Ceremony, I will decide on a faction; I will decide the rest of my life; I will decide to stay with my family or abandon them. "No," I say. "The tests don't have to change our choices." "Right." She smiles. "Let's go eat breakfast." "Thank you. For cutting my hair." She kisses my cheek and slides the panel over the mirror. I think my mother could be beautiful, in a different world. Her body is thin beneath the gray robe. She has high cheekbones and long eyelashes, and when she lets her hair down at night, it hangs in waves over her shoulders. But she must hide that beauty in Abnegation. We walk together to the kitchen. On these mornings when my brother makes breakfast, and my father's hand skims my hair as he reads the

Algorithm 2 Cross-excerpt Span Ratio

Require: Test book B with ordered excerpts $P = \{p_1, \ldots, p_n\}$, corresponding instructions $I = \{i_1, \ldots, i_n\}$, finetuned model M, minimum match length k
Ensure: Cross-excerpt ratio $\in [0, 1]$

$\mathcal{S} \leftarrow \emptyset$ ▷ Collection of (span, source) pairs
for each excerpt p_j with instruction i_j **do**
  for $t = 1$ to 100 **do**
    $g \leftarrow M(i_j)$
    Find all contiguous word matches $\geq k$ between g and B
    Add each match as $(span,\ p_j)$ to $\mathcal{S}$
Remove any span that is fully contained within a larger span
Deduplicate: collect the set of distinct source excerpt per unique span
for each unique span s **do**
  $target(s) \leftarrow$ the excerpt in B where s is located
  Mark s as *cross-excerpt* if any source $\neq target(s)$
return fraction of unique spans marked cross-excerpt

Best API match olmo-3-0625-32b-think

ai2-llm/pretraining-data/sources/cc_all_dressed/all_dressed_v3/weborganizer_ft/dclm_plus2_vigintiles/data/education_and_jobs/vigintile_0018/shard_00000404.jsonl.zst

[...] Our faction allows me to stand in front of it on the second day of every third month, the day my mother cuts my hair. I sit on the stool and my mother stands behind me with the scissors, trimming. The strands fall on the floor in a dull, blond ring. When she finishes, she pulls my hair away from my face and twists it into a knot. I note how calm she looks and how focused she is. She is well-practiced in the art of losing herself. I can't say the same of myself. I sneak a look at my reflection when she isn't paying attention—not for the sake of vanity, but out of curiosity. A lot can happen to a person's appearance in three months. In my reflection, I see a narrow face, wide, round eyes, and a long, thin nose—I still look like a little girl, though sometime in the last few months I turned sixteen. The other factions celebrate birthdays, but we don't. It would be self-indulgent. "There," she says when she pins the knot in place. Her eyes catch mine in the mirror. It is too late to look away, but instead of scolding me, she smiles at our reflection. I frown a little. Why doesn't she reprimand me for staring at myself? "So today is the day," she says. "Yes," I reply. "Are you nervous?" I stare into my own eyes for a moment. Today is the day of the aptitude test that will show me which of the five factions I belong in. And tomorrow, at the Choosing Ceremony, I will decide on a faction; I will decide the rest of my life; I will decide to stay with my family or abandon them. "No," I say. "The tests don't have to change our choices." "Right." She smiles. "Let's go eat breakfast." "Thank you. For cutting my hair." She kisses my cheek and slides the panel over the mirror. I think my mother could be beautiful, in a different world. Veronica Roth (Divergent (Divergent, #1)) [...]

C.2 Cross-excerpt spans

Cross-excerpt examples Section 5 shows that finetuned models frequently generate verbatim content from excerpts other than the one prompted. We quantify this with a cross-excerpt ratio for each model, as shown in Algorithm 2.

We also show representative examples of this behavior. For each example, we show the target excerpt (where the verbatim text originates in the book), the source excerpt (whose plot summary was used as the prompt), and the model's generation. Cross-excerpt spans are highlighted in yellow. We select examples across three books and models: The Remains of the Day (DeepSeek-V3.1), The Year of Magical Thinking (Gemini-2.5-Pro), and Midnight's Children (GPT-4o).

The Remains of the Day – DeepSeek-V3.1

Target excerpt: (excerpt id: 35)

Original Content:
I hope you will agree that in these two instances I have cited from his career – both of which I have had corroborated and believe to be accurate – my father not only manifests, but comes close to being the personification itself, of what the Hayes Society terms 'dignity in keeping with his position'. If one considers the difference between my father at such moments and a figure such as Mr Jack Neighbours even with the best of his technical flourishes, I believe one may begin to distinguish what it is that separates a 'great' butler from a merely competent one. We may now understand better, too, why my father was so fond of the story of the butler who failed to panic on discovering a tiger under the dining table; it was because he knew instinctively that somewhere in this story lay the kernel of what true 'dignity' is. And let me now posit this: 'dignity' has to do crucially with a butler's ability not to abandon the professional being he inhabits. Lesser butlers will abandon their professional being for the private one at the least provocation. For such persons, being a butler is like playing some pantomime role; a small push, a slight stumble, and the façade will drop off to reveal the actor underneath. The great butlers are great by virtue of their ability to inhabit their professional role and inhabit it to the utmost; they will not be shaken out by external events, however surprising, alarming or vexing. They wear their professionalism as a decent gentleman will wear his suit: he will not let ruffians or circumstance tear it off him in the public gaze; he will discard it when, and only when, he wills to do so, and this will invariably be when he is entirely alone. It is, as I say, a matter of 'dignity'.

Source excerpt: (excerpt id: 27)
Summary:
The excerpt is written predominantly in the first person from the perspective of the narrator, who engages in intellectual debates on the nature of 'dignity' with a character named Mr. Graham. The narrator disagrees with Mr. Graham's analogy that compares dignity to a woman's inherent beauty, suggesting instead that dignity is an attribute that can be cultivated over a butler's career, exemplified by figures like Mr. Marshall. The narrator recalls evenings spent in discussion with Mr. Graham, forming strong personal beliefs about dignity that continue to persist. The narrator uses examples of renowned butlers including Mr. Marshall, Mr. Lane, Mr. Henderson, and notably his own father, arguing his father embodied dignity at the height of his career at Loughborough House. This reflection highlights the narrator's contemplation on dignity as a quality achieved through self-training and professional experience.

Original Content:
Of course, this merely begs the further question: of what is 'dignity' comprised? And it was on this point that the likes of Mr Graham and I had some of our most interesting debates. Mr Graham would always take the view that this 'dignity' was something like a woman's beauty and it was thus pointless to attempt to analyse it. I, on the other hand, held the opinion that to draw such a parallel tended to demean the 'dignity' of the likes of Mr Marshall. Moreover, my main objection to Mr Graham's analogy was the implication that this 'dignity' was something one possessed or did not by a fluke of nature; and if one did not self-evidently have it, to strive after it would be as futile as an ugly woman trying to make herself beautiful. Now while I would accept that the majority of butlers may well discover ultimately that they do not have the capacity for it, I believe strongly that this 'dignity' is something one can meaningfully strive for throughout one's career. Those 'great' butlers like Mr Marshall who have it, I am sure, acquired it over many years of self-training and the careful absorbing of experience. In my view, then, it was rather defeatist from a vocational standpoint to adopt a stance like Mr Graham's. In any case, for all Mr Graham's scepticism, I can remember he and I spending many evenings trying to put our fingers on the constitution of this 'dignity'. We never came to any agreement, but I can say for my part that I developed fairly firm ideas of my own on the matter during the course of such discussions, and they are by and large the beliefs I still hold today. I would like, if I may, to try and say here what I think this 'dignity' to be. You will not dispute, I presume, that Mr Marshall of Charleville House and Mr Lane of Bridewood have been the two great butlers of recent times. Perhaps you might be persuaded that Mr Henderson of Branbury Castle also falls into this rare category. But you may think me

merely biased if I say that my own father could in many ways be considered to rank with such men, and that his career is the one I have always scrutinized for a definition of 'dignity'. Yet it is my firm conviction that at the peak of his career at Loughborough House, my father was indeed the embodiment of 'dignity'.

Model Generation:
I remember Mr Graham dismissing my view and suggesting that 'dignity' was something one possessed as a birthright – something one could not acquire, no matter how hard one tried. I recall him comparing 'dignity' to a beautiful woman's beauty; it was something one had or did not have, and the butlers who had it were, by virtue of this fact, in a different league from those of us who did not. I disagreed and argued that 'dignity' was something one could meaningfully strive for throughout one's career, and that among those of us who had devoted our lives to service, there were those who would be remembered as a 'great' butler – such as Mr Marshall or Mr Lane – while there were others who, though perhaps highly competent, would not leave a lasting impression on anyone, even on those they had served for decades. It was a quality, I argued, that could be acquired, but only after years of training and experience; a quality that set apart the great butlers from those who were merely good. I spent many evenings putting forth this view – just as, I am sure, I spent others arguing its opposite – but the fact is, the opinion I have just attributed to myself is one that has remained largely unchanged in my mind until this day. 'Dignity' has to do crucially with a butler's ability not to abandon the professional being he inhabits. Lesser butlers will abandon their professional being for the private one at the least provocation. For such persons, being a butler is like playing some pantomime role; a small push, a slight stumble, and the façade will drop off to reveal the actor underneath. The great butlers are great by virtue of their ability to inhabit their professional role and inhabit it to the utmost; they will not be shaken out by external events, however surprising, alarming or vexing. They wear their professionalism as a decent gentleman will wear his suit: he will not let ruffians or circumstance tear it off him in the public gaze; he will discard it when, and only when, he wills to do so, and this will invariably be when he is entirely alone. It is, as I say, a matter of 'dignity'. Of course, Mr Marshall, Mr Lane, Mr Henderson had it. But so for instance did my own father. Whatever one may say about his career, there can be no doubt that at the peak of it, around the time he achieved Loughborough House, my father certainly had 'dignity'.

The Year of Magical Thinking – Gemini-2.5-Pro

Target excerpt: (excerpt id: 3)

Original Content:
Nine months and five days ago, at approximately nine o'clock on the evening of December 30, 2003, my husband, John Gregory Dunne, appeared to (or did) experience, at the table where he and I had just sat down to dinner in the living room of our apartment in New York, a sudden massive coronary event that caused his death. Our only child, Quintana, had been for the previous five nights unconscious in an intensive care unit at Beth Israel Medical Center's Singer Division, at that time a hospital on East End Avenue (it closed in August 2004) more commonly known as "Beth Israel North" or "the old Doctors' Hospital," where what had seemed a case of December flu sufficiently severe to take her to an emergency room on Christmas morning had exploded into pneumonia and septic shock. This is my attempt to make sense of the period that followed, weeks and then months that cut loose any fixed idea I had ever had about death, about illness, about probability and luck, about good fortune and bad, about marriage and children and memory, about grief, about the ways in which people do and do not deal with the fact that life ends, about the shallowness of sanity, about life itself. I have been a writer my entire life. As a writer, even as a child, long before what I wrote began to be published, I developed a sense that meaning itself was resident in the rhythms of words and sentences and paragraphs, a technique for withholding whatever it was I thought or believed behind an increasingly impenetrable polish. The way I write is who I am, or have become, yet this is a case in which I wish I had instead of words and their rhythms a cutting room, equipped with an Avid, a digital editing system on which I could touch a key and collapse the sequence of time, show you simultaneously all the frames of

memory that come to me now, let you pick the takes, the marginally different expressions, the variant readings of the same lines. This is a case in which I need more than words to find the meaning. This is a case in which I need whatever it is I think or believe to be penetrable, if only for myself. We had seen Quintana in the sixth-floor ICU at Beth Israel North. We had come home. We had discussed whether to go out for dinner or eat in. I said I would build a fire, we could eat in. I built the fire, I started dinner, I asked John if he wanted a drink.

Source excerpt: (excerpt id: 106)
Summary:
The excerpt is written in the first-person voice and reflects on both a previous disdain for Caitlin Thomas's book, "Leftover Life to Kill," and a traumatic medical episode involving the narrator's husband. The narrator recalls initially judging Caitlin Thomas for her perceived self-pity but then reflects on their own cognitive deficits and emotional struggles during a medical emergency. The progression changes from reflections on the past to a detailed chronological account of a medical emergency involving her husband, who is described as having suffered cardiac arrest on December 30, 2003. Various medical interventions are recorded, but he is ultimately pronounced dead at 10:18 p.m. The narrator's memory issues amid the crisis further complicate the situation. The details convey a mix of personal regret and the stark realities of witnessing a loved one's sudden death.

Original Content:
I remember despising the book Dylan Thomas's widow Caitlin wrote after her husband's death, Leftover Life to Kill. I remember being dismissive of, even censorious about, her "self-pity," her "whining," her "dwelling on it." Leftover Life to Kill was published in 1957. I was twenty-two years old. Time is the school in which we learn. A doctor to whom I mentioned this shrugged, as if I had told him a familiar story. Either he said that such "cognitive deficits" could be associated with stress or he said that such cognitive deficits could be associated with grief. It was a mark of those cognitive deficits that within seconds after he said it I had no idea which he had said. According to the hospital's Emergency Department Nursing Documentation Sheet, the Emergency Medical Services call was received at 9:15 p.m. on the evening of December 30, 2003. According to the log kept by the doormen the ambulance arrived five minutes later, at 9:20 p.m. During the next forty-five minutes, according to the Nursing Documentation Sheet, the following medications were given, by either direct injection or IV infusion: atropine (times three), epinephrine (times three), vasopressin (40 units), amiodarone (300 mg), high-dose epinephrine (3 mg), and high-dose epinephrine again (5 mg). According to the same documentation the patient was intubated at the scene. I have no memory of an intubation. This may be an error on the part of whoever did the documentation, or it may be another cognitive deficit. According to the log kept by the doormen the ambulance left for the hospital at 10:05 p.m. According to the Emergency Department Nursing Documentation Sheet the patient was received for triage at 10:10 p.m. He was described as asystolic and apneic. There was no palpable pulse. There was no pulse via sonography. The mental status was unresponsive. The skin color was pale. The Glasgow Coma Scale rating was 3, the lowest rating possible, indicating that eye, verbal, and motor responses were all absent. Lacerations were seen on the right forehead and the bridge of the nose. Both pupils were fixed and dilated. "Lividity" was noted. According to the Emergency Department Physician's Record the patient was seen at 10:15 p.m. The physician's notation ended: "Cardiac arrest. DOA—likely massive M.I. Pronounced 10:18 p.m." According to the Nursing Flow Chart the IV was removed and the patient extubated at 10:20 p.m. At 10:30 p.m. the notation was "wife at bedside—George, soc. worker, at bedside with wife."

Model Generation:
I had not read it in forty years. What I had seen in it then was self-pity, and a maddening lack of clarity. Yet what I see in it now is an early case study in a syndrome of cognitive deficit increasingly noted in people who have experienced loss. One day I was talking to the social worker at New York University Hospital who had become my confidant. I told her that I could not seem to think, could not finish sentences, could not manage simple tasks. I told her that I had just that morning gone to the bank to make a deposit and had been unable to remember in which of my two accounts I should deposit the check. I had repeated the situation to the social worker as if it were a comical vignette, a "story" I might have told about myself before December 30, but in fact it had not been remotely comical. There in the bank I had begun to weep. This account will not be the one I began. This account will be about how I

tried to come to terms with the year that followed. This is my attempt to make sense of the period that followed, weeks and then months that cut loose any fixed idea I had ever had about death, about illness, about probability and luck, about good fortune and bad, about marriage and children and memory, about grief, about the ways in which people do and do not deal with the fact that life ends, about the shallowness of sanity, about life itself. I have been a writer my entire life. As a writer, even as a child, long before what I wrote began to be published, I developed a sense that meaning itself was resident in the rhythms of words and sentences and paragraphs, a technique for withholding whatever it was I thought or believed behind an increasingly impenetrable polish. The way I write is who I am, or have become, yet this is a case in which I wish I had instead of words and their rhythms a cutting room, equipped with an Avid, a digital editing system on which I could touch a key and collapse the sequence of time, show you simultaneously all the frames of memory that come to me now, let you pick the takes, expedite the transitions. At nine-thirty on the evening of December 30, 2003, my husband, John Gregory Dunne, appeared to (or did) experience, at the table where he was reading, a sudden massive coronary event that caused his death. One minute he was talking to me about Jack London. The next minute he fell silent.

Midnight's Children – GPT-4o

Target excerpt: (excerpt id: 433)

Original Content:
What my discovery of unfairness (smelling of onions) had begun, my invisible rage completed. Wrath enabled me to survive the soft siren temptations of invisibility; anger made me determined, after I was released from vanishment in the shadow of a Friday Mosque, to begin, from that moment forth, to choose my own, undestined future. And there, in the silence of graveyard-reeking isolation, I heard the long-ago voice of the virginal Mary Pereira, singing: Anything you want to be, you kin be, You kin be just what-all you want. Tonight, as I recall my rage, I remain perfectly calm; the Widow drained anger out of me along with everything else. Remembering my basket-born rebellion against inevitability, I even permit myself a wry, understanding smile. "Boys," I mutter tolerantly across the years to Saleem-at-twenty-four, "will be boys." In the Widows' Hostel, I was taught, harshly, once-and-for-all, the lesson of No Escape; now, seated hunched over paper in a pool of Anglepoised light, I no longer want to be anything except what who I am. Who what am I? My answer: I am the sum total of everything that went before me, of all I have been seen done, of everything done-to-me. I am everyone everything whose being-in-the-world affected was affected by mine. I am anything that happens after I've gone which would not have happened if I had not come. Nor am I particularly exceptional in this matter; each "I," every one of the now-six-hundred-million-plus of us, contains a similar multitude. I repeat for the last time: to understand me, you'll have to swallow a world. Although now, as the pouring-out of what-was-inside-me nears an end; as cracks widen within—I can hear and feel the rip tear crunch—I begin to grow thinner, translucent almost; there isn't much of me left, and soon there will be nothing at all. Six hundred million specks of dust, and all transparent, invisible as glass … But then I was angry. Glandular hyper-activity in a wicker amphora: eccrine and apocrine glands poured forth sweat and stink, as if I were trying to shed my fate through my pores; and, in fairness to my wrath, I must record that it claimed one instant achievement—that when I tumbled out of the basket of invisibility into the shadow of the mosque, I had been rescued by rebellion from the abstraction of numbness; as I bumped out on to the dirt of the magicians' ghetto, silver spittoon in hand, I realized that I had begun, once again, to feel. Some afflictions, at least, are capable of being conquered.

Source excerpt Example 1: (excerpt id: 37)
Summary:
In this excerpt, the narrator, speaking in the first person, is being urged by Padma, a woman who is both critical and caring, to maintain a linear storytelling style. Padma chides the narrator for the slow pace of his narrative, suggesting that he'll take forever to reach the story of his birth. Despite her nonchalant demeanor and complaints, Padma is deeply engrossed in his story. She has become so invested that she has settled into the narrator's life, preparing his

food and spending nights in his workspace. The narrator reflects on the interconnectedness of events and people, suggesting that stories and lives intermingle like flavors in cooking. While Padma argues for a more straightforward storytelling approach, her presence and influence are seeping into the narrator's life. The narrator acknowledges Padma's generosity and patience in sticking by him despite his inability to engage with her romantically. In essence, the excerpt explores the dynamic relationship between the narrator and Padma, while highlighting themes of storytelling, human connection, and frustration.

Original Content:
But here is Padma at my elbow, bullying me back into the world of linear narrative, the universe of what-happened-next: "At this rate," Padma complains, "you'll be two hundred years old before you manage to tell about your birth." She is affecting nonchalance, jutting a careless hip in my general direction, but doesn't fool me. I know now that she is, despite all her protestations, hooked. No doubt about it: my story has her by the throat, so that all at once she's stopped nagging me to go home, to take more baths, to change my vinegar-stained clothes, to abandon even for a moment this darkling pickle-factory where the smells of spices are forever frothing in the air … now my dung goddess simply makes up a cot in the corner of this office and prepares my food on two blackened gas-rings, only interrupting my Anglepoise-lit writing to expostulate, "You better get a move on or you'll die before you get yourself born." Fighting down the proper pride of the successful storyteller, I attempt to educate her. "Things—even people—have a way of leaking into each other," I explain, "like flavors when you cook. Ilse Lubin's suicide, for example, leaked into old Aadam and sat there in a puddle until he saw God. Likewise," I intone earnestly, "the past has dripped into me … so we can't ignore it … " Her shrug, which does pleasantly wavy things to her chest, cuts me off. "To me it's a crazy way of telling your life-story," she cries, "if you can't even get to where your father met your mother." … And certainly Padma is leaking into me. As history pours out of my fissured body, my lotus is quietly dripping in, with her down-to-earthery, and her paradoxical superstition, her contradictory love of the fabulous—so it's appropriate that I'm about to tell the story of the death of Mian Abdullah. The doomed Hummingbird: a legend of our times. … And Padma is a generous woman, because she stays by me in these last days, although I can't do much for her. That's right—and once again, it's a fitting thing to mention before I launch into the tale of Nadir Khan—I am unmanned. Despite Padma's many and varied gifts and ministrations, I can't leak into her, not even when she puts her left foot on my right, winds her right leg around my waist, inclines her head up toward mine and makes cooing noises; not even when she whispers in my ear, "So now that the writery is done, let's see if we can make your other pencil work!"; despite everything she tries, I cannot hit her spittoon.

Model Generation:
PADMA is stirring again. (Don't worry: I'm not pushing her away.) Since my head is full of fog, do what I will, I must be content with this slow one-page-after-another philosophy…no, not content; that's the wrong word. I will be content when my story has been told; when I have managed to say, I am the sum total of everything that went before me, of all I have been seen done, of everything done-to-me. I am everyone everything whose being-in-the-world affected was affected by mine. I am anything that happens after I've gone which would not have happened if I had not come. Nor am I particularly exceptional in this matter; each "I," every one of the now-six-hundred-million-plus of us, contains a similar multitude. I repeat for the last time: to understand me, you'll have to swallow a world. One hundred years of newspapers and cinemas! ET CETERA! What's real and what's true aren't necessarily the same. My head relearned this fact, but it may need recapitulation. Who what am I? My answer: I am the sum total of everything that went before me, of all I have been seen done, of everything done-to-me. I am everyone everything whose being-in-the-world affected was affected by mine. I am anything that happens after I've gone which would not have happened if I had not come. Nor am I particularly exceptional in this matter; each "I," every one of the now-six-hundred-million-plus of us, contains a similar multitude. I repeat for the last time: to understand me, you'll have to swallow a world. Once, on the back of an envelope, I wrote: How we FREEDOMTEMPTERS would saunter down Connaught Place in evening, calling out Telugu private parts names full of fire extinguisher-juice? Will they stop making aeroplanecoloured stone? Who let off this enormous stinkbomb? Favourite rebuttals: LIKE I TOLD YOU SO! LIKE I WAS SAYING! and LF/AIB. Each…did something, to our FEARLESSNESS like the song of that great singer who died in such mysterious circumstances— Shree? Pantoo? Slide down a little, in the maybe chair! What Apples caled

the platelet. Entharness AND in Carstairs ? & so on. Anyhow dear what was there about them teeth? Such things enter my thoughts. "Get to the point," Padma says, "or what's a heaven for?" So now, patiently (for is not patience a virtue?) she says, "What is so precious to find out? Commissioner sahib gave somebody a break once?" "And that was shameful for him?" she wants to know; I ignore her (for the moment) and press on. And, "If you held off on your birth at the rate you're going, you'd finish on the second of June a hundred and thirty years from now!"

Source excerpt Example 2: (excerpt id: 115)
Summary:
In this excerpt, written in the first-person voice, the narrator—presumably Saleem—is recounting an intricate web of personal and familial history that intertwines with significant cultural and historical elements. Padma, another character, is present and prompts the narrator to begin telling their story. The narrator reflects on their lineage, describing the inheritance of traits and experiences from ancestors, including a grandfather who took an oath and encountered a boatman with prophetic insights. This familial history is rich with symbolic events and characters, such as landowners, lady wrestlers, and the mystical perforated sheet. The passage conveys a sense of destiny shaped by these inherited elements—ranging from supernatural prophecies and mythical objects to familial traits of skepticism and credulity—culminating in the narrator's realization of viewing their life and its meanings in fragmented parts. The scene portrays the weight of the past as a legacy mingling with the narrator's identity and future.

Original Content:
But today, Padma heard Mountbatten's ticktock … English-made, it beats with relentless accuracy. And now the factory is empty; fumes linger, but the vats are still; and I've kept my word. Dressed up to the nines, I greet Padma as she rushes to my desk, flounces down on the floor beside me, commands: "Begin." I give a little satisfied smile; feel the children of midnight queueing up in my head, pushing and jostling like Koli fishwives; I tell them to wait, it won't be long now; I clear my throat, give my pen a little shake; and start. Thirty-two years before the transfer of power, my grandfather bumped his nose against Kashmiri earth. There were rubies and diamonds. There was the ice of the future, waiting beneath the water's skin. There was an oath: not to bow down before god or man. The oath created a hole, which would temporarily be filled by a woman behind a perforated sheet. A boatman who had once prophesied dynasties lurking in my grandfather's nose ferried him angrily across a lake. There were blind landowners and lady wrestlers. And there was a sheet in a gloomy room. On that day, my inheritance began to form—the blue of Kashmiri sky which dripped into my grandfather's eyes; the long sufferings of my great-grandmother which would become the forbearance of my own mother and the late steeliness of Naseem Aziz; my great-grandfather's gift of conversing with birds which would descend through meandering bloodlines into the veins of my sister the Brass Monkey; the conflict between grandpaternal scepticism and grandmaternal credulity; and above all the ghostly essence of that perforated sheet, which doomed my mother to learn to love a man in segments, and which condemned me to see my own life—its meanings, its structures—in fragments also; so that by the time I understood it, it was far too late. Years ticking away—and my inheritance grows, because now I have the mythical golden teeth of the boatman Tai, and his brandy bottle which foretold my father's alcoholic djinns; I have Ilse Lubin for suicide and pickled snakes for virility; I have Tai-for-changelessness opposed to Aadam-for-progress; and I have, too, the odors of the unwashed boatman which drove my grandparents south, and made Bombay a possibility.

Model Generation:
I am handed down; meanwhile Padma and I stare at each other across the soap-stained folds of my bedspread, three children gazing through the window enclosing them in the perforated sheet. "Begin at the beginning," the hump-backed crone advises me; "Go on," she urges, "don't be so slow! Start the stories—there are so many—instead of bothering with all this chewed-up mess." Who what am I? My answer: I am the sum total of everything that went before me, of all I have been seen done, of everything done-to-me. I am everyone everything whose being-in-the-world affected was affected by mine. I am anything that happens after I've gone which would not have happened if I had not come. Nor am I particularly exceptional in this matter; each "I," every one of the now-six-hundred-million-plus of us, contains a similar multitude. I repeat for the last time: to understand me, you'll have to swallow a world. One

		N	Mean Rank	Top 10%
Overall	Observed	13,263	**0.746**	**42.5%**
	Random baseline	13,263	0.495	9.7%
By model	GPT-4o	9,228	0.743	42.4%
	Gemini-2.5-Pro	3,655	0.758	44.3%
	DeepSeek-V3.1	1,427	0.821	56.7%
By setting	Within-author	1,220	0.746	44.8%
	Cross-author	12,043	0.746	42.3%
By distance	1–5 excerpts	3,886	0.888	72.1%
	6–20 excerpts	2,155	0.754	41.8%
	21–50 excerpts	2,215	0.681	28.8%
	51+ excerpts	5,007	0.660	26.0%

Table 9: **Semantic similarity analysis of cross-excerpt retrieval.** For each cross-excerpt span, we rank the triggered excerpt among all excerpts in the book by cosine similarity to the prompt. A mean rank of 0.5 and top-10% rate of 10% correspond to random retrieval. Triggered excerpts are 4.4× more likely than random to fall in the top 10%, consistent across models, experiment settings, and excerpt distances.

> further note, then; if you are to judge me, you must bear in mind that the ending to my story is (to date) a trifle premature; in there, too, depths and depths. Clues. For this reason I must give warning of my tendency to "explain" everything ; until history firmly reserves its verdict on me, I am obliged to offer my opinions. Unbiased eyes would do me a great service. Well then: "If you really want to go the whole hog," Padma says threateningly, "you'll have to start before you were born't." Genesis is, after all, not a mere matter of ideology or theological phil-horror-sophies; it's also a feminine business, and cannot begin until the woman (not to be contrasted with a picture of some huge cosmic guy in the sky) is duly big with child; and my poor unaware mummy Amina only got into that any-old-iron pot of frying meat because of a fisherman's stinking boat and another oath which resulted in our having to leave Kashmir. So now, by the jagged and unusual line of Kashmir's fate-twisted mountains, I command the story to begin. The water genie, still trapped by glass, is quarrelling with the clock-tower man outside old Hangman. Meanwhile, beneath the surface of Lake Dal in the heart of Kashmir, a battle is continuing between land and water; and the boatman Tai's face has become granite.

Cross-excerpt span semantic similarity analysis To test whether cross-excerpt retrieval is driven by semantic similarity, we measure how the triggered excerpt ranks among all excerpts in the same book by similarity to the prompt. For each cross-excerpt span, we take the plot summary and compute its cosine similarity to every excerpt in the book using OpenAI `text-embedding-3-small` (OpenAI, 2024). We then compute the rank percentile of the actual triggered excerpt: a value of 1.0 means it is the most similar excerpt in the book, while 0.5 is the expected value under random retrieval. As a baseline, we sample one random excerpt per pair from the same book and compute its rank under the same similarity distribution. We deduplicate cross-excerpt pairs by (book, source excerpt, target excerpt), counting each semantic relationship once regardless of how many models produce it.

Table 9 reports the results. Overall, triggered excerpts rank at the 74.6th percentile in semantic similarity to the prompt, compared to 49.5th for the random baseline, and 42.5% fall in the top 10% most similar excerpts, which is 4.4× the random rate of 9.7%. The effect is consistent across all three finetuned models as they all show strong semantic targeting, and near-identical results for within-author and cross-author (both 0.746) settings confirm that the retrieval structure is independent of whether the model was finetuned on the same author. To rule out positional proximity as an alternative explanation, we stratify by excerpt distance. While nearby excerpts show the strongest effect (0.888 mean rank for distance 1–5), excerpts more than 50 positions apart—where surface-level overlap is minimal—still rank at 0.660 with a top-10% rate of 26.0%, well above the random baseline.

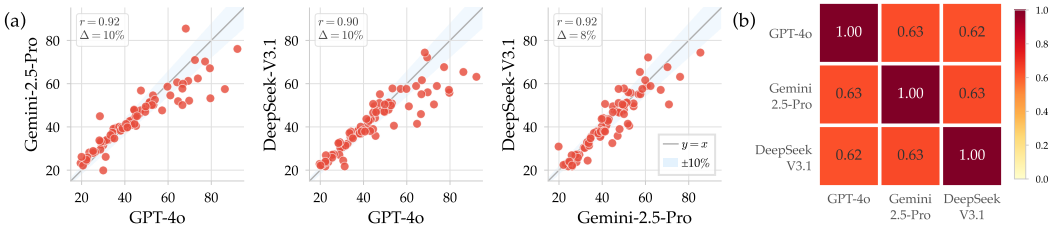


Figure 11: **Different models show strikingly similar memorization patterns. (a)** Per-book bmc@5 scatter plots for each pair of finetuned models. Each point is one book; the diagonal line marks perfect agreement, with the shaded band indicating $\pm 10\%$. All pairs show strong correlation ($r \geq 0.90$) and small deviations ($\Delta \leq 10\%$), indicating that models consistently agree on which books are more or less extractable. **(b)** Average word-level Jaccard similarity across all books. The pairwise similarity reaches 90-97% of each model's own self-agreement ceiling (0.650-0.689), meaning the three models memorize nearly identical regions within each book despite different architectures and providers.

C.3 Cross-model memorization agreement

Figure 11 provides detailed evidence of cross-model memorization agreement. Panel (a) shows per-book bmc@5 scatter plots for each pair of finetuned models. Each point represents one book; the diagonal line marks perfect agreement, with the shaded band indicating $\pm 10\%$. All pairs show strong correlation ($r \geq 0.90$) and small deviations ($\Delta \leq 10\%$), indicating that models consistently agree on which books are more or less extractable. Panel (b) reports average word-level Jaccard similarity across all books. For each model, computing bmc@5 produces a binary mask over word positions in the book. To interpret the pairwise Jaccard values, we establish two reference points: a random baseline from shuffled masks, and an upper bound from each model's self-agreement (split-half over 100 generations per excerpt), representing the agreement ceiling given sampling stochasticity. Pairwise cross-model similarities reach 90-97% of self-agreement (0.650-0.689), far above the random baseline, meaning that nearly all content extractable from one model is also extractable from the others.